ELEC2134 Lecture Notes

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Part I
Transform Methods

1 Topic 1: Periodic Signals and Fourier Series

1.1 Waveforms and Periodic Functions

1.1.1 Waveform Fundamentals

Definition 1.1.

- A waveform describes the shape of a signal as a function of time.
- A signal is a function that varies with time.
- A unit step signal stays at 0 until a certain time, then instantly becomes 1.
- A ramp signal increases linearly with time, starting from 0.

The period (T) is the time taken for a waveform to complete one full cycle and start repeating

$$\text{frequency } f = \frac{1}{T}\text{Hertz (Hz)}.$$

Hertz is cycles per second. The units of period is time.

Example 1.2. Some common frequencies.

Frequency (f)	Period ($f = \frac{1}{T}$)
1 kHz	$1 \times 10^{-3}s = 1ms$
1 MHz	$1 \times 10^{-6}s = 1\mu s$
1 GHz	$1 \times 10^{-9}s = 1ns$
1 THz	$1 \times 10^{-12}s = 1ps$

The peak amplitude of a wave is the distance between the x axis and the highest point of the wave. The peak-to-peak amplitude is two times this amplitude.

Periodic Waveforms (or Functions) 1.3.
A periodic waveform or function is any waveform or function that satisfies

$$x(t) = x(t + T),$$

where t is the time variable, and T is the period. In general,

$$x(t) = x(t + nT), \forall n \in \mathbb{Z}.$$

1.1.2 Sinusoid waveforms

A sinusoidal waveform is a type of period waveform, and can be described using the following equation

$$x(t) = A\sin(2\pi ft),$$

where $x(t)$ is the instantaneous amplitude, A is the peak amplitude, and f is the frequency in Hz. It can also be described in terms of period with

$$x(t) = A\sin\left(\frac{2\pi}{T}t\right),$$

where T is the period. This can be simplified in terms of angular frequency $\omega = 2\pi f = \frac{2\pi}{T}$ radians/sec as

$$x(t) = A\sin(\omega t).$$

1.1.3 Other Types of Waveforms

Square Waveform

A square waveform is used mostly in digital electronic circuits for clock and timing control signals. It has very steep vertical transitions and flat top and bottoms.

It has pulse width $\tau = \frac{T}{2}$ and frequency $f = \frac{1}{T}$.

Rectangular Waveform

A rectangular waveform is similar to a square wave, however the duration of the high level (pulse width) is shorter than half the period. This is described by the duty cycle, which is the amount of time a signal is ON compared to the total time of one cycle.

Triangular Waveform

A triangular waveform is a non-sinusoidal waveform that linearly oscillates between a positive and negative peak. They are usually comprised of many sine or cosine waves with different frequencies and amplitudes.

Sawtooth Waveform

A sawtooth waveform is a periodic waveform whose shape resembles the teeth of a sawblade. It is similar to a triangular waveform, however the decay is more similar to a square wave's.

1.1.4 Harmonic Components

A harmonic is a multiple of the fundamental angular frequency of a waveform. The fundamental frequency is the frequency as defined above, giving the harmonics to be

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T} \quad \Rightarrow \quad w_n = nw_0, f_n = nf_0,$$

where n is a positive integer.

The average value of a waveform is the mean level of the signal over one complete cycle. This is the overall DC component of the waveform. For a periodic waveform $x(t)$ with period T, this is

$$V_{avg} = \frac{1}{T}\int_0^T x(t)dt.$$

1.2 Fourier Series

1.2.1 Frequency Domain Representation

The frequency domain representation of a waveform shows how a time-domain (i.e. amplitude as a function of time) signal is distributed across different frequencies, effective decomposing a signal into it's component frequencies.

Example 1.1. What does the phase-domain of the sinusoid $v(t)$ below look like?

$$v(t) = \frac{4}{\pi}\sin(\omega_0 t) + \frac{4}{3\pi}\sin(3\omega_0 t) + \frac{4}{5\pi}\sin(5\omega_0 t).$$

Answer: This would be a plot with three peaks; one at ω_0, one at $3\omega_0$, and one at $5\omega_0$. The first one has a height of $4/\pi$, the second 43π, and the last $4/5\pi$.

1.2.2 Fourier Series

This process of breaking down a signal is called Fourier decomposition, resulting in a Fourier series. This is a sum of sinusoidal components with different frequencies and amplitudes. The main application of a Fourier series is to determine the harmonic components that are present in a complex periodic waveform.

Any periodic waveform with a period of T seconds can be represented by an infinite series of harmonically related sinusoids. The representation is known as the trigonometric Fourier series.

$$x(t) = A_0 + \sum_{i=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t).$$

The fundamental frequency of $x(t)$ is $\omega_0 = 2\pi f_0 = \frac{2\pi}{T}$ radians/sec. The integer multiples of ω_0 are known as harmonic frequencies of $x(t)$. The Fourier coefficients A_0, A_n and B_n are calculated directly from the signal $x(t)$ using the following equations

$$\begin{aligned}
A_0 &= \frac{1}{T}\int_{-T/2}^{T/2} x(t)dt \\
\omega_0 &= 2\pi f_0 = \frac{2\pi}{r} \text{ radians/sec} \\
A_n &= \frac{2}{T}\int_{-T/2}^{T/2} x(t)\cos(n\omega_n t)dt \\
B_n &= \frac{2}{T}\int_{-T/2}^{T/2} x(t)\sin(n\omega_n t)dt.
\end{aligned}$$

Fourier Series of an Even Periodic Function

A function is defined as even if $x(t) = x(-t)$. For these functions/waveforms, the expressions for the Fourier coefficients reduce to

$$A_0 = \frac{2}{T}\int_0^{T/2} x(t)dt, \quad A_n = \frac{4}{T}\int_0^{T/2} x(t)\cos(n\omega_0 t)dt, \quad B_n = 0.$$

The reason that the B coefficients disappear is because the periodic waveform is even, so there are no terms from the odd sine function.

Fourier Series of an Odd Periodic Function

A function is defined as odd if $x(t) = -x(t)$. For these function/waveforms, the expressions reduce to

$$A_0 = A_n = 0 \text{ for } n > 0, \quad B_n = \frac{4}{T}\int_0^{T/2} x(t)\sin(n\omega_0 t)dt$$

1.2.3 Reconstructing a Waveform

To reconstruct (or synthesise) a waveform/function from a set of Fourier coefficients, simply plug them back into the above expression for Fourier series along with their index.

1.3 Alternative Trigonometric Form

In the frequency domain, the amplitude of each frequency component is determined by A_n and B_n, with contributions from both the sine and cosine terms. We would now like to express the signal in an alternative form in which the amplitude of each frequency components is obtained using a single cosine function with a phase angle

$$x(t) = K_0 = \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n).$$

This is the amplitude-phase representation. We are able to find the phase angle ϕ_n with

$$\phi_n = \tan^{-1}\left(-\frac{B_n}{A_n}\right) = -\tan^{-1}\left(\frac{B_n}{A_n}\right).$$

With this phase angle, we are also able to express the Fourier series using a complex number with $K_n \angle \phi_n = A_n - jB_n$.

1.4 Complex Form of the Fourier Series

The cosine-sine and amplitude-phase Fourier series from above are all trigonometric Fourier series. Another alternative Fourier series representations uses exponential functions instead of trigonometric functions. This form is therefore called the complex form of the (or exponential) Fourier series, and is given by

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad C_n = \frac{1}{T}\int_0^T x(t)e^{-jn\omega_0 t}dt,$$

where $\omega_0 = \frac{2\pi}{T}$. The DC component is then given by C_0, and C_n can be found with

$$C_n = \frac{1}{2}(A_n - jB_n), \quad |C_{-n}| = |C_n| = \frac{1}{2}\sqrt{A_n^2 + A_n^2}.$$

- For even symmetry: $C_n = \frac{2}{T}\int_0^{T/2} x(t)\cos(n\omega_0 t)dt$
- For odd symmetry: $C_n = -\frac{2j}{T}\int_0^{T/2} x(t)\sin(n\omega_0 t)dt$

1.5 Effect of Shifting

Shifting a periodic function along either the vertical or the horizontal axis will generally affect its Fourier coefficients. These are also affected by the presence of certain types of functional symmetries in $f(t)$, particularly even-, odd-, and half-wave symmetry because some coefficient sets vanish altogether.

Shifting a periodic function $f(t)$ along either the vertical or the horizontal axis affects its amplitude-phase Fourier series. It is apparent that shifting $f(t)$ up or down the vertical axis changes only the average value K_0, leaving the amplitude K_n and phase angle ϕ_n of each harmonics component unaffected.

1.6 Half-Wave Symmetry

An important type of symmetry is the half-wave symmetry. A periodic function is half-wave symmetric if

$$x(t) = -x\left(t - \frac{T}{2}\right).$$

Shifting a half-wave symmetric wave or function by half a period then inverting it yields the original function. If a function has half-wave symmetric, then both A_n and B_n are 0 for even values of n.

1.6.1 Odd Half-Wave Symmetry

A periodic signal has odd half-wave symmetry is both of the following are true:

$$x(t) = -x\left(t - \frac{T}{2}\right), \quad \text{and} \quad x(t) = -x(-t).$$

For a function with odd half-wave symmetry, the coefficients are given by

$$A_n = \begin{cases} \frac{4}{T}\int_0^{T/2} x(t)\cos(n\omega_0 t)dt & \text{for n odd} \\ 0 & \text{for n even} \end{cases}$$

$$B_n = \begin{cases} \frac{4}{T}\int_0^{T/2} x(t)\sin(n\omega_0 t)dt & \text{for n odd} \\ 0 & \text{for n even} \end{cases}$$

For odd half-wave symmetry

- All cosine terms are 0 $A_n = 0$.
- All even harmonics are 0.
- Only odd sine harmonics appear in the Fourier series.

1.6.2 Even Half-Wave Symmetry

A periodic signal has even half-wave symmetry is both of the following are true:

$$x(t) = -x\left(t - \frac{T}{2}\right), \quad \text{and} \quad x(t) = x(-t).$$

For a function with even half-wave symmetry, the coefficients are given by

$$A_n = \begin{cases} \frac{4}{T}\int_0^{T/2} x(t)\cos(n\omega_0 t)dt & \text{for n odd} \\ 0 & \text{for n even} \end{cases}$$

$$B_n = \begin{cases} \frac{4}{T}\int_0^{T/2} x(t)\sin(n\omega_0 t)dt & \text{for n odd} \\ 0 & \text{for n even} \end{cases}$$

For even half-wave symmetry

- All sine terms are 0 $B_n = 0$.
- All sine harmonics are 0.
- Only odd cosine harmonics appear in the Fourier series.

1.7 Examples

1.7.1 Waveforms

Example 1.1. Sum these two sine waves

$$f(t) = 10\sin(200t)$$
$$g(t) = 8\sin(200t)$$

Answer: These sine waves have the same period $2\pi f = 200$, $f = \frac{200}{2\pi}Hz = 31.83Hz$.

2 Topic 2: Aperiodic Signals and Fourier Transforms

2.1 Periodic and Aperiodic Signals

A periodic signal repeats itself after some period T, and can be represented by

$$x(t) = x(t + nT), \quad \forall n \in \mathbb{Z}.$$

The frequency components of a periodic signal are analysed using a Fourier series. An aperiodic signal has no repeating pattern, and we assume it's period is infinite. Most practical signals are aperiodic. Since the fundamental frequency is inversely proportionate to the period, so as the period increases indefinitely the individual spectral components merge into a continuous spectrum and the fundamental frequency becomes vanishingly small.

As a consequence,the frequency $n\omega_0$ of each harmonic component becomes the continuous frequency variable ω, the line spacing ω_0 becomes the infinitesimal $d\omega$, and the operation of summation becomes the operation of integration.

Consider the exponential Fourier series. We have

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad C_n = \frac{1}{T}\int_{-T/2}^{T/2} x(t)e^{-jn\omega_0 t}dt.$$

This then gives the full expression for $x(t)$ to be

$$x(t) = \sum_{n=-\infty}^{\infty} \left\{ \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-j\omega_0 t} dt \right\} e^{jn\omega_0 t},$$

and the inverse to be

$$f(t) = \sum_{n=-\infty}^{\infty} \left\{ \frac{1}{2\pi} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \right\} e^{jn\omega_0 t} \omega_0.$$

We can turn this Fourier series into a Fourier transform by taking the limit of $T \to \infty$, replacing the frequency with the continuous frequency, the line spacing with $d\omega$, and integrating instead of summing. This gives the Fourier transform (FT) of $x(t)$ (denote $X(\omega)$ or $X(j\omega)$) to be

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt,$$

and the inverse Fourier transform (IFT) to be

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega.$$

Note that $X(\omega)$ is a complex-valued function, and $x(t)$ is normally a real valued signal.

2.1.1 Fourier Series to Fourier Transform

- For small T, the signal repeats frequently in time and its frequency spectrum consists of discrete spectral lines that are widely spread. As T increases and approaches ∞, the pulses move further apart in time, the fundamental frequency decreases, and the spacing between spectral lines becomes smaller
- As a consequence, more frequency components appear within a given frequency range, and the discrete spectrum begins to resemble a continuous one
- In the limit, the periodic signal becomes an aperiodic single pulse, the spectral lines merge into a continuous spectrum, and the Fourier series representation naturally transitions to the Fourier transform.

2.2 Fourier Transform Properties

Let $f(x)$ and $h(x)$ represent integrable functions satisfying

$$\int_{-\infty}^{\infty} |f(x)|\, dx < \infty.$$

We denote the Fourier transforms of these functions as $F(\xi)$ and $H(\xi)$, respectively.

2.2.1 Basic Properties

The Fourier transform has the following basic properties

- **Linearity:** $af(x) + bh(x) \overset{\mathcal{F}}{\Longleftrightarrow} aF(\xi) + bH(\xi), \quad a, b \in \mathbb{C}$
- **Time shifting:** $f(x - x_0) \overset{\mathcal{F}}{\Longleftrightarrow} e^{-i2\pi x_0 \xi} F(\xi), \quad x_0 \in \mathbb{R}.$
- **Frequency shifting:** $e^{i2\pi \xi_0 x} f(x) \overset{\mathcal{F}}{\Longleftrightarrow} F(\xi - \xi_0), \quad \xi_0 \in \mathbb{R}$
- **Time scaling:** $f(ax) \overset{\mathcal{F}}{\Longleftrightarrow} \frac{1}{|a|} F\left(\frac{\xi}{a}\right), \quad a \neq 0$. Note: the case $a = -1$ leads to the time-reversal property: $f(-x) \overset{\mathcal{F}}{\Longleftrightarrow} F(-\xi)$

2.2.2 Differentiation

Suppose $f(x)$ is differentiable almost everywhere, and both f and its derivative f' are integrable. The Fourier transform is then given by

$$F'(\xi) = \mathcal{F}\left\{\frac{d}{dx}f(x)\right\} = i2\pi\xi F(\xi).$$

We also have differentiation in time to be

$$\frac{d}{dt}f(x) \overset{\mathcal{F}}{\Longleftrightarrow} j\xi F(\xi),$$

which is multiplying the frequency by $j\xi$. Similarly, taking the integrate w.r.t time gives

$$\int_{-\infty}^{t} f(x)dx \overset{\mathcal{F}}{\Longleftrightarrow} \frac{F(\xi)}{j\xi}.$$

And for frequency we have

$$-jtf(x) \overset{\mathcal{F}}{\Longleftrightarrow} \frac{d}{d\xi}F(\xi),$$

which is effectively multiplication in time by jt.

2.2.3 Convolution

Convolution describes how the output of a system depends on the input signal and the system's impulse response. Taking the same functions $f(x)$, $g(x)$ from before (along with their transforms), we get an output signal $y(x)$ given by

$$y(x) = f(x) * g(x) = \int_{-\infty}^{\infty} f(\tau)g(x-\tau)d\tau,$$

where $*$ denotes the convolution operation, then

$$Y(\xi) = F(\xi)G(\xi).$$

Convolution in the time domain corresponds to multiplication in the frequency domain. Note that when the input is an impulse $f(x) = \delta(x)$, then

$$Y(\xi) = G(\xi),$$

since $X(\xi) = 1$.

2.2.4 Modulation

Amplitude modulation is the process of varying the amplitude of a sinusoidal carrier. If the modulating signal is denoted $x(t)$, the modulated carrier becomes

$$y(t) = x(T)\cos(\omega_0 t).$$

The amplitude spectrum of this carrier is one-half the amplitude spectrum of $x(t)$ shifted to be centred at $\pm\omega_0$

$$\mathcal{F}\left\{x(t)\cos(\omega_0 t)\right\} = \frac{1}{2}X(\omega-\omega_0) + \frac{1}{2}X(\omega+\omega_0).$$

3 Topic 3: Laplace Transforms of Signals and Circuits

3.1 Introduction to Laplace Transform

A Laplace transform can be thought of as a generalisation of Fourier transforms that transforms differential equations into algebraic equations. It is also able to handle initial conditions, transients, and steady-states responses.

It is an integral transformation that maps a function $f(t)$ from the time domain to the s-domain. It is defined as

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t)e^{-st}dt,$$

where s is a complex variable and is given by $s = \sigma + j\omega$, σ is the exponential damping factor, and ω is the frequency. Note that the Fourier transform is a special case when $\sigma = 0$,

- $s = j\omega$ - sinusoidal AC signals (Fourier/frequency axis)
- $s = \sigma$ - for exponential signals
- $s = \sigma + j\omega$ - for exponential sinusoidal signals.

3.2 Laplace Transform Properties

Let $f(t)$ and $g(t)$ be two functions with respective Laplace transforms $F(s)$ and $G(s)$

$$f(t) = \mathcal{L}^{-1}\{F(s)\}, \quad g(t) = \mathcal{L}^{-1}\{G(s)\}.$$

3.2.1 Basic Properties

The Laplace transform has the following basic properties (note: the $u(t-a)$ function is present to ensure the function doesn't include negative time):

- **Linearity**: $af(t) + bg(t) \overset{\mathcal{L}}{\Longleftrightarrow} aF(s) + bG(s)$
- **Time scaling**: $f(at) \overset{\mathcal{L}}{\Longleftrightarrow} \frac{1}{a}F\left(\frac{s}{a}\right)$
- **Time shifting**: $f(t-a)u(t-a) \overset{\mathcal{L}}{\Longleftrightarrow} e^{-as}F(s)$
- **Frequency shifting**: $e^{-as}f(t)u(t) \overset{\mathcal{L}}{\Longleftrightarrow} F(s+a)$
- **First time derivative**: $\frac{df(t)}{dt} \overset{\mathcal{L}}{\Longleftrightarrow} sF(s) - f(^-)$
- **Second time derivative**: $\frac{d^2f(t)}{dt^2} \overset{\mathcal{L}}{\Longleftrightarrow} s^2F(s) - sf(0^-) - f'(0^-)$
- **General time derivative**: $\frac{d^n}{dt^n}f(t) \overset{\mathcal{L}}{\Longleftrightarrow} s^nF(s) + \sum_{i=1}^n \left[-s^{n-k}f^{k-1}(0^-)\right]$
- **Time integration**: $\int_0^t f(\tau)d\tau \overset{\mathcal{L}}{\Longleftrightarrow} \frac{1}{s}F(s)$
- **Differentiation in s-domain**: $tf(t) \overset{\mathcal{L}}{\Longleftrightarrow} -\frac{dF(s)}{ds}$
- **Differentiation in s-domain (general)**: $t^nf(t) \overset{\mathcal{L}}{\Longleftrightarrow} (-1)^nF^{(n)}(s)$
- **Frequency Integration**: $\frac{f(t)}{t} \overset{\mathcal{L}}{\Longleftrightarrow} \int_s^\infty F(s)ds$

3.2.2 Initial and Final Value Theorem

The initial and final values of a function can be directly found from its Laplace transform. We have the initial value to be

$$f(0) = \lim_{s\to\infty} sF(s),$$

and final value as

$$f(\infty) = \lim_{s\to 0} sF(s).$$

Note that the final value theorem is only applicable when the system has a definite final value. For rational $F(s)$, the poles of $sF(s)$ must lie in the open left-half place (aside from the simple origin case).

Example 3.1. Find the final value of the function

$$f(t) = e^{-2t}\sin(5t)u(t).$$

Answer: First, we find the Laplace transform of this function to be

$$F(s) = \mathcal{L}\{f(t)\} = \frac{5}{(s+2)^2 + 4s^29}.$$

Taking the limit gives

$$\begin{aligned} f(\infty) &= \lim_{s\to 0} sF(s) \\ &= \lim_{s\to 0} \frac{5s}{s^2+4s+29} \\ &= 0. \end{aligned}$$

3.3 Inverse Laplace Transform

The inverse Laplace transform is very similar to the formula for the inverse Fourier transform. We have

$$f(t) = \mathcal{L}^{-1}\{F(s)\}(t) = \frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT} e^{st}F(s)ds,$$

where $\gamma = \mathfrak{Re}(s)$, such that γ is greater than the real part of all singularities (discontinuities) of F and F is bounded on the line. In the special case where all singularities lie in the open left half-place, or F is an entire function (such as a polynomial, exponential, finite sum, etc), then γ can be set to 0 and the above integral is identical to the inverse Fourier transform.

The typical way to find the inverse Laplace transform is to simply using the properties from above, along with a table of basic/common Laplace transform pairs, to find the inverse. This usually also requires decomposing $F(s)$ into partial fractions

3.3.1 Partial Fraction Decomposition

The partial fraction decomposition of a rational fraction of the form

$$\frac{f(x)}{g(x)},$$

where $f(x)$, $g(x)$ are polynomials and $g(x) \neq 0$, is the expression of the rational fractions as

$$\frac{f(x)}{g(x)} = p(x) + \sum_k \frac{f_j(x)}{g_j(x)},$$

where $p(x)$ is a polynomial, and for each j, the denominator $g_j(x)$ is a power of an irreducible polynomial (i.e. not factorisable into polynomials of positive degrees), and the numerator $f_j(x)$ is a polynomial of a

smaller degree than the degree of the irreducible polynomial. Note that when doing this by hand, we just consider "square-free" polynomials rather than irreducible polynomials.

There are two methods for finding partial fractions:

- **Residue Method**: "Cover" the factor $(x-p)$, then evaluate the rest at $x=p$. See Example 3.6 for an example of this.
- **Algebraic Method**: Equate $f(x)$ to a coefficient corresponding to one of the factors multiplied by the rest of $g(x)$. Then, the coefficients can be found by solving for each of them. See Example 3.6 for an example of this.

3.4 Laplace-Domain Circuit Models

Laplace transform is particularly useful for circuit analysis by converting to the s-domain of circuit elements. Circuit elements can be transformed into impedances, which is similar to phasor impedances from ELEC1111.

3.4.1 Resistor

For resistors, all values remain the same in both the time domain and the s-domain.

$$v(t) = Ri(t) \overset{\mathcal{L}}{\Longleftrightarrow} V(s) = RI(s).$$

3.4.2 Inductors

In the time-domain, we have voltage to be inductances multiplied by the change in current, and the current to be the total change in voltage divided by the inductance,

$$v(t) = L\frac{di(t)}{dt}, \quad i(t) = \frac{1}{L}\int_0^t v(\tau)d\tau + i(0),$$

where $i(0)$ is the current at $t=0$. Taking the Laplace transform of this, we get

$$\begin{aligned} v(t) = L\frac{di(t)}{dt} &\overset{\mathcal{L}}{\Longleftrightarrow} V(s) = sLI(s) - Li(0) \\ i(t) = \frac{1}{L}\int_0^t v(\tau)d\tau + i(0) &\overset{\mathcal{L}}{\Longleftrightarrow} I(s) = \frac{V(s)}{sL} + \frac{i(0)}{s}. \end{aligned}$$

3.4.3 Capacitors

In the time-domain, the voltage is given by the total change in current divided by the capacitance, and the current to be the capacitance multiplied by the change in voltage.

$$v(t) = \frac{1}{C}\int_0^t i(\tau)d\tau + v(0), \quad i(t) = C\frac{dv(t)}{dt}.$$

Taking the Laplace transform of this, we get

$$\begin{aligned} v(t) = \frac{1}{C}\int_0^t i(\tau)d\tau + v(0) &\overset{\mathcal{L}}{\Longleftrightarrow} V(s) = \frac{I(s)}{sC} + \frac{v(0)}{s} \\ i(t) = c\frac{dv(t)}{dt} &\overset{\mathcal{L}}{\Longleftrightarrow} I(s) = sCV(s) - Cv(0). \end{aligned}$$

3.4.4 Initial Conditions

In the above equations, the initial conditions represented a voltage or current source on the circuit, indicating the circuit started with a certain current/voltage. If we assume no initial conditions, we get

Table 1: Zero initial conditions in Laplace-domain circuit models

Voltage/Current Relation	Impedance	Admittance
$V(s) = RI(s)$	$Z(s) = \frac{V(s)}{I(s)} = R$	$Y(s) = \frac{I(s)}{V(s)} = \frac{1}{R}$
$V(s) = sLI(s)$	$Z(s) = \frac{V(s)}{I(s)} = sL$	$Y(s) = \frac{I(s)}{V(s)} = \frac{1}{sL}$
$V(s) = \frac{I(s)}{sC}$	$Z(s) = \frac{V(s)}{I(s)} = \frac{1}{sC}$	$Y(s) = \frac{I(s)}{V(s)} = sC$

3.5 Transfer Functions

A transfer function describes how a signal is changed whilst passing through a system. The transfer function $H(s)$ is defined as the ratio of the output response $Y(s)$ to input excitation $X(s)$, assuming all initial conditions are zero

$$H(s) = \frac{Y(s)}{X(s)}, \quad Y(s) = H(s) \cdot X(s),$$

where $H(s)$ is the transfer function, and $X(s)$ and $Y(s)$ are the input and output, respectively. If both $X(s)$ and $Y(s)$ are voltages, then it represents a voltage gain. If both variables are currents, it represents a current gain. This gives four common types of transfer function

- **Voltage Gain**: $H(s) = \frac{V_o(s)}{V_i(s)}$
- **Current Gain**: $H(s) = \frac{I_o(s)}{I_i(s)}$
- **Impedance**: $H(s) = \frac{V(s)}{I(s)}$
- **Admittance**: $H(s) = \frac{I(s)}{V(s)}$

A single circuit may have several possible transfer function, depending on the chosen input and output variables. Note that gain is dimensionless, whereas impedance and admittance have units.

3.5.1 Unit Impulse Response

To determine the transfer function $H(s)$ of a system, both in the input and output functions must be known. If the input $X(s)$ and the transfer function are known, the output is obtained as

$$Y(s) = H(s) \cdot X(s).$$

Taking the inverse Laplace transform,

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}.$$

If the input is a unit response, $x(t) = \delta(t)$, then

$$X(s) = 1,$$

and therefore $Y(s) = H(s)$, which also means

$$y(t) = h(t),$$

where

$$h(t) = \mathcal{L}^{-1}\{H(s)\}.$$

The function $h(t)$ is called the unit impulse response. It represents the time-domain response of the system to a unit impulse input. Therefore, the transfer function $H(s)$ is the Laplace transform of the unit impulse response of the system.

3.6 Examples

Example 3.1. Find the inverse Laplace transform of

$$F(s) = \frac{s^2 + 12}{s(s+2)(s+3)}$$

using both the Residue method and the algebraic method for partial fraction decomposition.

Answer: First, we need to find the partial fractions

$$\frac{s^2 + 12}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3}.$$

Algebraic Method: Multiplying both sides by the denominator, we get

$$s^2 + 12 = A(s^2 + 5s + 6) + B(s^2 + 3s) + C(s^2 + 2s).$$

Equating the coefficients of like powers, we get

$$\begin{aligned}
\text{Constant}: \quad & 12 = 6A && \Rightarrow A = 2 \\
s: \quad & 0 = 5A + 3B + 2C && \Rightarrow 3B + 2C = -10 \\
s^2: \quad & 1 = A + B + C && \Rightarrow B + C = -1 \\
& \therefore A = 2, B = -8, C = 7.
\end{aligned}$$

Residual Method: To find each coefficient, we just multiply the denominator of the respective fraction by $F(s)$. This gives the following set of equations

$$A = sF(s)\Big|_{s=0} \tag{1}$$

$$B = (s+2)F(s)\Big|_{s=-2} \tag{2}$$

$$C = (s+3)F(s)\Big|_{s=-3}. \tag{3}$$

Evaluating these gives

$$\begin{aligned}
(1) \quad A &= \frac{s^2 + 12}{(s+2)(s+3)}\Big|_{s=0} \\
&= \frac{12}{2 \times 3} = 2 \\
(2) \quad B &= \frac{s^2 + 12}{s(s+3)}\Big|_{s=-2} \\
&= \frac{12}{-2 \times 1} = -8 \\
(3) \quad C &= \frac{s^2 + 12}{s(s+2)} \\
&= \frac{12}{-3 \times -1} = 7.
\end{aligned}$$

We can see that these give the same result, however the second method was easier. We now have our function to be in the form

$$F(s) = \frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3},$$

which we can find the inverse of using linearity and a table of common pairs to get

$$\begin{aligned}
f(t) &= \mathcal{L}^{-1}\left(\frac{2}{s}\right) - \mathcal{L}^{-1}\left(\frac{8}{s+2}\right) + \mathcal{L}^{-1}\left(\frac{7}{s+3}\right) \\
&= 2u(t) - 8e^{-2t} + 7e^{-3t}, \quad t \geq 0
\end{aligned}$$

Example 3.2. Use the Laplace transform to solve the following differential equation

$$\frac{d^2v(t)}{dt^2} + 6\frac{dv(t)}{dt} + 8v(t) = 2u(t),$$

given $v(0) = 1$, $v'(0) = -2$.

Answer:

Taking the Laplace transform of each term gives

$$\left[s^2V(s) - sv(0) - v'(0)\right] + 6\left[sV(s) - v(0)\right] + 8V(s) = \frac{2}{s}.$$

Substituting $v(0) = 1$ and $v'(0) = -2$, we obtain

$$\begin{aligned}(s^2 + 6s + 8)V(s) &= s + 4 + \frac{2}{s} \\ &= \frac{s^2 + 4s + 2}{s} \Rightarrow V(s) \qquad = \frac{1/4}{s} + \frac{1/2}{s+2} + \frac{1/4}{s+4}.\end{aligned}$$

Taking the inverse Laplace transform, we obtain

$$v(t) = \frac{1}{4}(1 + 2e^{-2t} + e^{-4t})u(t).$$

Example 3.3. Use the Laplace transform to solve the following differential equation

$$\frac{dy}{dt} + 5y(t) + 6\int_0^t y(\tau)d\tau = u(t),$$

given $y(0) = 2$.

Answer:

Taking the Laplace transform of each term in the given differential equation yields

$$[sY(s) - y(0)] + 5Y(s) + \frac{6}{s}Y(s) = \frac{1}{s}.$$

Substituting $y(0) = 2$, we obtain

$$\begin{aligned}Y(s)(s^2 + 5s + 6) &= 1 + 2s \\ Y(s) &= \frac{2s+1}{(s+2)(s+3)} \\ &= -\frac{3}{s+2} + \frac{5}{s+3}.\end{aligned}$$

Taking the inverse Laplace transform, we get

$$y(t) = (-3e^{-2t} + 5e^{-3t})u(t).$$

Part II
System Analysis

4 Topic 4: System Analysis

An analogue system is a device that operates on a continuous time signal $x(t)$, called the input, and produces another continuous time signal $y(t)$, called the output (or response), according to a well-defined rule.

4.1 s-Plane

Usually, it is best to represent the complex frequency $s = \sigma + j\omega$ graphically in the complex plane, called the $s - plane$. The horizontal axis σ represents the real part of s, i.e. the exponential damping factor, and the vertical axis $j\omega$ represents the imaginary part of s, i.e. the sinusoidal frequency ω.

The Fourier transform can be obtained from the Laplace transform by setting $\sigma = 0$. Thus, if the region of convergence for the Laplace transform includes the imaginary axis, then,

$$\text{FT} : H(j\omega) = H(s)|_{s=j\omega}.$$

In this plane, $\sigma = 0$ corresponds to the imaginary axis. Therefore, the Fourier transform is obtained by evaluating the Laplace transform along the imaginary axis.

4.2 Poles and Zeros

Poles: When the denominator $X(s)$ of a rational transfer function $H(s) = \frac{Y(s)}{X(s)}$ of a system is set equal to zero, it is known as the characteristic equation. The roots of this equation determine the character of the time response of the system. These roots are called the poles of the system, and they are where the transfer function becomes infinite.
Zeros: Similarly, the roots of the numerator polynomial $Y(s)$ of a transfer function are called the zeros of the system, and correspond to where the transfer function is zero.

We denote the location of zeros in the s-plane with the symbol 0, and the location of poles with the symbol $\times$.

4.3 System Stability

4.3.1 Real Pole(s)

If all of a systems pole lie in the left-half of the $s - plane$, then that system is said to be stable. They correspond to exponentially decaying impulse responses. For a causal continuous-time linear-time-invariant (LTI) system, Bounded-Input Bounded-Output (BIBO) stability requires all poles of $H(s)$ to lie strictly in the open left-half plane, i.e. $\mathfrak{Re}(p) < 0$.

> **Note 4.1.** A system is causal if its output at time t only depends on present and past inputs, not future inputs. For an LTI system, this means its impulse satisfies
>
> $$H(t) = 0, \quad \forall t < 0.$$
>
> This means that if a unit impulse response contains $u(t)$, it is inherently causal.

If the poles instead lie in the right half-plane, then they correspond to systems with exponentially growing impulse responses, and are called unstable as they do not converge.

4.3.2 Complex Conjugate Poles

Consider a transfer function with a pair of complex conjugate poles at

$$H(s) = \frac{1}{(s-p_1)(s-p_3)}$$
$$= \frac{1}{(s-(\sigma_a+j\omega_a))(s-(\sigma_a-j\omega_a))} \qquad = \frac{A}{s-\sigma_a-j\omega_a} + \frac{B}{s-\sigma_a+j\omega_a}.$$

By evaluating the constants A and B, we obtain

$$\begin{aligned} H(s) &= \frac{\frac{1}{2j\omega_a}}{s-\sigma_a-j\omega_a} + \frac{-\frac{1}{2j\omega_a}}{s-\sigma_a+j\omega_a} \\ \Rightarrow h(t) &= \mathcal{L}^{-1}\{H(s)\} \\ &= \frac{1}{2j\omega_a}\left[e^{(\sigma_a+j\omega_a)t} - e^{(\sigma_a-j\omega_a)t}\right] \\ &= \frac{1}{\omega_a}\cdot e^{\sigma_a t}\left[\frac{e^{j\omega_a t} - e^{-j\omega_a t}}{2j}\right] \\ &= \frac{1}{\omega_a} e^{\sigma_a t}\sin(\omega_a t). \end{aligned}$$

This is a impulse response $h(t)$ of a second-order system with complex conjugate poles. The real part σ_a determines the decay rate, while the imaginary part ω_a determines the oscillation frequency. Thus, the impulse response depends strongly on the pole locations.

If the real part of a pole is kept constant and the imaginary part increases from ω_a to ω_p, the envelope of the impulse response $h(t)$ remains the same, while the oscillation frequency increases.

If the imaginary part is held constant and the real part becomes more negative (move further into the left-half plane), the oscillation frequency remains unchanged while the decay rate increases. Notice also that if the pole lies on the real axis (i.e. $\omega_a = 0$), then there is no oscillation in the impulse response. If the poles lies on the imaginary axis ($\sigma = 0$), there is no growth or decay in the impulse response, it is a pure oscillation. If a system has a pole with a positive real part ($\sigma > 0$), its impulse response grows exponentially. Such a system is unstable.

A stable and causal system is a system that has all its poles in the left half-plane. A persistent undamped oscillation is not BIBO stable.

4.4 Analysis of a System Without Zeros

4.4.1 First-Order System

A first-order system without zeros can be described by the transfer function

$$H(s) = \frac{a}{s+a},$$

with the pole at $-a$. If the input is a unit step function, where $X(s) = \frac{1}{s}$, the Laplace transform of the step response is

$$Y(s) = X(s)\cdot H(s) = \frac{1}{s}\frac{a}{s+a} = \frac{1}{s} - \frac{1}{s+a}.$$

Taking the inverse transform of $Y(s)$, the step response is given by

$$y(t) = 1 - e^{-at}.$$

We can define a time constant T to be

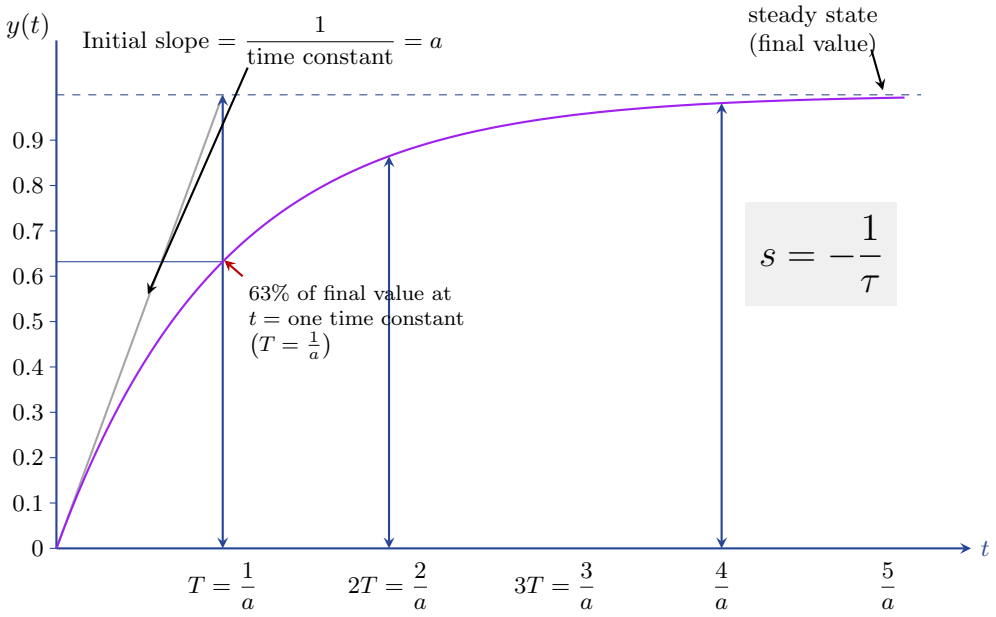
$$T = \frac{1}{a},$$

which we can plug into this step response

$$y(t)\Big|_{t=\frac{1}{a}} = 1 - e^{-a\left(\frac{1}{a}\right)} = 0.63,$$

which means T define the amount of time it takes for the step response to rise to 63% of its final value, and has the unit seconds. The reciprocal of the time constant has the units $\frac{1}{\text{sec}}$, or frequency. We can then call a the cutoff frequency.

The time constant is related to the transient response, which is the speed at which the system responds to a step input. Since the pole of the transfer function is at $s = -a$, we can say the pole is located at the negative of the reciprocal of the time constant, and the further the pole from the imaginary axis the faster the transient response.



4.4.2 Second-Order System

In comparison to a first-order system, a second-order system exhibits a wide range of responses. The general case for a second-order transfer function is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{a}{s^2 + bs + a}.$$

By assigning appropriate values to a and b, we can see the different transient responses. The term a is simply a multiplier that can take any value. If the input $X(s)$ is a unit step input $X(s) = \frac{1}{s}$, then the output $Y(s) = X(S) \cdot H(s) = \frac{1}{s}\frac{a}{s^2+bs+a}$,

$$\therefore y(t) = \mathcal{L}^{-1}\left[\frac{1}{s}\frac{a}{s^2 + bs + a}\right].$$

We can rewrite the transfer function as

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

which we can use to describe different types of systems

Case	Damping	s-plane	Unit step response
$a = b$, $\zeta > 1$	Overdamped	Two real poles	
$a > b$, $0 < \zeta < 1$	Underdamped	Two complex poles	
$b = 0$, $\zeta = 0$	Undamped	Two imaginary poles	
$b = \frac{1}{2}\sqrt{a}$, $\zeta = 1$	Critically damped	Two repeated real poles	

4.5 Cascade and Parallel Structures

A linear system can be characterised by its transfer function $H(s)$, where its input $X(s)$ and output $Y(s)$. When the transfer function of the overall system $H(s)$ is the product of two transfer functions, then it is called cascade form (can be thought of as in series)

$$H(s) = H_1(s) \cdot H_2(s).$$

When two transfer functions $H_3(s)$ and $H_4(s)$ appear in parallel, then the overall transfer function is given by

$$H(s) = H_3(s) + H_4(s).$$

4.6 Frequency Response

The steady-state response $\sigma = 0$ of a linear system to a sinusoidal input is obtained from the expression for $H(s)$ by replacing s with $j\omega$. The function $H(j\omega) = H(\omega)$ is called the frequency response function. The modulus of $H(j\omega)$ gives the magnitude of the sinusoidal response $|H(j\omega)| = |H(\omega)$. The angle of $H(j\omega)$ gives the phase of the response $\theta(\omega)$, relative to the input sinusoid.

4.6.1 Decibels

The decibel is a logarithmic measurement unit for sound levels, but also is important in electronics and signal processing. It represents ratios such as power, sound pressure, voltage, intensity, etc. using a logarithmic scale

$$\text{decibels } dB = 20 \log_{10} \frac{v_{out}}{v_{in}} = 10 \log_{10} \left(\frac{v_{out}}{v_{in}} \right)^2 = 10 \log_{10} \frac{P_{out}}{P_{in}}.$$

A difference of 3 dB in a reading represents a doubling in power, and -3 dB represents half power. Each successive multiple of ten increases the order of magnitude by 10.

4.7 Low-Pass Filter

For a low-pass filter or system, we have

$$H(s) = \frac{a}{s + a},$$

with the magnitude

$$|H(j\omega)| = \frac{a}{\sqrt{a^2 + \omega^2}}.$$

The cutoff occurs when

$$|H(j\omega)| = \frac{1}{\sqrt{2}},$$

which gives

$$\omega = a = \omega_c,$$

where ω_c is called the cutoff frequency. The total bandwidth of the system is then

$$\text{BW} = a.$$

A first-order low-pass filter has the form

$$H(s) = \frac{\omega_c}{s + \omega_c}, \quad H(j\omega) = \frac{\omega_c}{j\omega + \omega_c}$$

The cutoff frequency can also be defined by when the magnitude drops off by 3 dB. For a second-order low-pass filter, this is just two low-pass filters cascaded together. Both first- and second-order low-pass filters have the same cutoff frequency, and each pole contributes approximately -20 dB/decade to the high-frequency asymptotic slope. A first-order filter rolls off at 20 dB/decade beyond the cutoff frequency, and a second-order filter rolls at at 40 dB/decade asymptotically for low-pass systems with two more poles than zeros, meaning it attenuates unwanted frequencies sharper than a first-order system.

4.8 Second-Order Low Pass Filter

A second-order low pass filter has the form

$$H(s) = \frac{\omega_0^2}{s^2 + b_0 s + \omega_0^2},$$

with bandwidth and cutoff frequency being given by

$$b_0 = \frac{\omega_0}{Q}, \quad \omega_0,$$

where Q is the quality factor. In the case were the real poles are distinct, the cut-off frequency ω_1 is determined by the dominate pole α_1.

When the poles are complex conjugates, it becomes a resonant low-pass filter. It has the same form as the above second-order low-pass filters.

4.9 Band-Pass Filter

A band-pass filter has two cutoff frequencies ω_u and ω_l which identify the pass-band (i.e. the region in which frequencies are minimally attenuated). The cutoff frequencies are defined as the frequencies for which the magnitude of the transfer function is $\frac{1}{\sqrt{2}}$ times the original magnitude. There are three important parameters in a band-pass filter.

The first important parameter in a band-pass filter is the centre frequency ω_o, which is defined as the frequency for which a circuit's transfer function is purely real. This is also known as the resonant frequency, and can be found as

$$\omega_o = \sqrt{\omega_u \omega_l}.$$

For band-pass filters, the magnitude of the transfer function is maximum at the centre frequency.

The second parameter is the bandwidth (BW), which is the width of the pass-band

$$BW = \omega_u - \omega_l.$$

The final parameter is the quality factor Q, which is the ratio of the centre frequency to the bandwidth

$$Q = \frac{\omega_o}{BW}.$$

This is a measure of the width of the pass-band, independent of its location on the frequency axis. It also describes the shape of the magnitude plot, independent of frequency.

4.9.1 Second-Order Band-Pass Filter

A second-order band-pass filter has the form

$$H(s) = \frac{b_0 s}{s^2 + b_0 s + \omega_0^2},$$

with b_0 is again the bandwidth $b_0 = \frac{\omega_2}{Q}$, and Q is the quality factor. Note the s in the numerator, this is not present in the resonant low-pass filters.

4.10 Notch and Band-Stop Filters

A notch filter suppresses a specific frequency (known as the notch frequency), ideally reducing its magnitude to zero. It has the form

$$H(s) = \frac{s^2 + \omega_z^2}{\omega_z^2},$$

where ω_z is the notch frequency. When complex poles are introduced to a notch filter, it becomes a band-stop filter

$$H(s) = \frac{s^2 + \omega_z^2}{s^2 + b_p s + \omega_p^2}$$

with b_p the bandwidth of the resonant pole, ω_p the resonant frequency of the pole, Q_p the quality factor of the pole, and ω_z the notch frequency. When the pole frequency $\omega_p = \omega_z$, we get a standard notch filter, depending on $b_p = \frac{\omega_p}{Q_p}$, the notch will be narrower or wider.

4.11 High-Pass Filter

A high-pass filter passes high frequency signals and attenuates low frequency signals

$$H(s) = \frac{s}{s + \omega_c}, \quad H(j\omega) = \frac{j\omega}{j\omega + \omega_c}.$$

This also blocks DC components, because $|H(j0)| = 0$. A second-order high-pass filter with real poles has the form

$$H(s) = \frac{s^2}{s^2 + b_1 s + \omega_1^2},$$

and similarly with complex poles it has the form

$$H(s) = \frac{s^2}{s^2 + b_2 s + \omega_2^2}.$$

5 Part 5: Bode Plot

5.1 Decibels

The ratio of any two values of the same quantity can be expressed in decibels. It is often useful to express the magnitudes of transfer functions in decibels. To convert a transfer function magnitude to decibels, we multiply the common logarithm (base 10) of the transfer function magnitude by 20.

$$|H(j\omega)|_{dB} = 20 \log_{10} \left| \frac{V_{out}(j\omega)}{V_{in}(j\omega)} \right|.$$

Similarly, the ratios of powers are converted to decibels by taking 10 times the logarithm of the ratio

$$dB = 10 \log_{10} \left| \frac{P_{out}}{P_{in}} \right|.$$

Note that the decibels equivalents are positive for magnitudes greater than unity, whereas the decibel equivalents are negative for magnitudes less than unity. An advantage of decibels is when plotting is that very small and very large magnitudes can be displayed clearly on a single plot. For cascaded systems, the product of the transfer functions becomes sums in decibels.

A decade is a range of frequencies for which the ratio of the highest frequency to the lowest is 10 (i.e. 2-20Hz is one decade, 50-5000Hz is two decades). An octave is a two-to-one change in frequency (i.e. 10-20Hz is one octave, 2-16Hz is three octaves). The number of decades between two frequencies is given by

$$\log_{10} \frac{f_2}{f_1}.$$

Similarly, the number of octaves between two frequencies is

$$\log_2 \frac{f_2}{f_1} = \frac{\log_{10} \frac{f_2}{f_1}}{\log_{10} 2}.$$

5.2 Bode Plot

A bode plot is abscissa (logarithmic frequency) vs magnitude (in dB) or degrees.

Part III
AC Circuit Analysis

6 AC Fundamentals

In alternating current (AC), the flow of electric charge periodically changes direction, whereas in direct current (DC), the electric charge only ever flows in a single direction. AC is more efficient and economical to transmit over long distances, and has the advantage of being able to step-up or step-down the voltage using a transformer.

6.1 Characteristics of Alternating Current and Voltage

For AC power systems, there are two important quantities,

- **Power loss**: the power losses P_L in a conductor are a product of the square of the current and resistance of the conductor
$$P_L = I^2 R.$$
When transmitting a fixed power through a power line, if the current is doubled, the power loss will be four times greater.
- **Power transmitted**: the power transmitted P_T is equal to the product of the current and the voltage
$$P_T = VI.$$
The same amount of power can be transmitted with a lower current by increasing the voltage, which leads to significantly more efficient transmission of power.

An alternating current or voltage can have different waveforms, but for the most part we will focus on sinusoidal waveforms, given by

$$v(t) = V_m \sin(2\pi f t) = V_m \sin\left(\frac{2\pi}{T} t\right),$$

where $v(t)$ is the instantaneous voltage at time t, V_m is the peak voltage, f is the frequency, and T is the period. Frequency is related to period by

$$f = \frac{1}{T} \text{ Hz},$$

but we can simplify both by expressing in angular velocity

$$\omega = 2\pi f = \frac{2\pi}{f} \text{ rad/s},$$

which we can use to write our sinusoid as

$$v(t) = V_m \sin \omega t.$$

6.2 Phase Angle

A sinusoidal waveform may or may not always start at $t = 0$ (i.e. have it's first 0 crossing at $t = 0$). A delay or advance in the waveform is known as the phase angle (ϕ), which is in degrees or radians. We can include this in our sinusoidal waveform as

$$v(t) = V_m \sin(\omega t \pm \phi).$$

Note that an advance looks like the waveform has been shifted to the left, however this is an advance as it reaches it's peak *before* a sinusoid with no phase angle.

6.2.1 Phase Difference

Suppose a waveform A is chosen as the reference waveform with the sinusoid given by

$$v(t) = V_m \sin(\omega t).$$

A waveform B is said to lead the waveform A by 45°if it has a phase angle of $\phi = +45°$,

$$v(t) = V_m \sin(\omega t + 45°).$$

A waveform C is said to lag the waveform A by 60°if it has a phase angle of $\phi = -60°$,

$$v(t) = V_m \sin(\omega t - 60°).$$

Note that a sinusoid can also be expressed as a cosine,

$$v(t) = V_m \sin(\omega t + 90°) = V_m \cos \omega t, \quad v(t) = V_m \cos(\omega t - 90°) = V_m \sin \omega t.$$

6.3 Measurements of AC Quantities

For a sinusoidal waveform, we can measure

- Peak value: the maximum value the sinusoid reaches
- Peak-to-peak value: the voltage difference between the minimum and maximum of the sinusoid
- Average value: the average value of the sinusoid, usually called the DC component. This is the offset from the horizontal axis.
- Root Mean Square (RMS) value: the measure of the heating effect of the sinusoid. It is the practical amount of power in a sinusoid.

6.3.1 Peak Value

The peak value of a sinusoid is known as the maximum voltage (or current) of the waveform, and occurs at two different positions in a cycle (one positive peak and one negative peak). The magnitude of these peaks are equal, and are 180° out of phase.

6.3.2 Peak-to-Peak Value

The peak-to-peak value of an AC waveform is the total height between opposite peaks, i.e. the difference between the magnitude of the positive peak and the magnitude of the negative peak.

6.3.3 Average Value

The average value of a sinusoid is the average of all its instantaneous values over a full cycle, given by

$$V_{avg} = \frac{1}{T}\int_0^T v(t)dt = \frac{V_m}{T\omega}\left[\cos(\omega t)\right]_{\frac{2\pi}{\omega}}^{0}.$$

Since the positive and negative alternations are (usually) identical, the average is zero.

6.3.4 Root Mean Square Value

The root mean square value (RMS) is a measure of the heating effect of a sine wave. It can be thought of as the equivalent value of DC that will product the same amount of power in a load as a given sine wave. It is given by

$$v_{rms} = \frac{V_m}{\sqrt{2}}, \quad i_{rms} = \frac{I_m}{\sqrt{2}}.$$

This is independent of the phase angle or frequency of sine wave, and depends only on the amplitude of the waveform. The I_{rms} of a periodic waveform $i(t)$ is defined as the current which produces the heat (i.e. power) in a given resistor R at the same average rate as a direct current I_{dc}

$$P_{average} = RI_{rms}^2 = RI_{dc}^2.$$

Consider a periodic current waveform. We can find I_{rms} as

$$\begin{aligned} P_{avg} &= \frac{1}{T}\int_0^T p(t)dt \\ &= \frac{1}{T}\int_0^T i^2(t)Rdt \\ &= \frac{R}{T}\int_0^T i^2(t)dt = RI_{rms}^2 \\ \therefore I_{rms}^2 &= \frac{1}{T}\int_0^T i^2(t)dt \\ I_{rms} &= \sqrt{\frac{1}{T}\int_0^T i^2(t)dt}. \end{aligned}$$

Similarly for voltage,

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt}.$$

For a waveform comprised of sinusoids of different frequencies, the RMS value is given by the sum of the squaer of the RMS values of each sinusoid. For an AC current $i(t)$ given by

$$i(t) = I_0 + \sum_k^n I_{m,k}\sin(\omega_k t),$$

we have the RMS value of $i(t)$ to be

$$I_{rms}^2 = I_0^2 + \sum_k^n I_{rms,k}^2.$$

6.4 Phasor Representation

A phasor is a line drawn from an origin that has both magnitude and angle. They are used in AC circuits to indicate time varying quantities of current and voltage. They have the form

$$\mathbf{V} = V_m\angle\phi,$$

where $\mathbf{V}$ is the phasor, V_m is the magnitude, and ϕ is the phase angle. It is a complex number that represents the amplitude and phase angle of a sinusoid. We can represent it in different forms

- Rectangular: $\mathbf{V} = V_m\cos\phi + jV_m\sin\phi$
- Polar: $\mathbf{V} = V_m\angle\phi$
- Exponential: $\mathbf{V} = V_m e^{j\phi}$

6.5 The Voltage-Current Relationship for Components

6.5.1 Resistor

Assume that the current through a resistor is given by

$$i(t) = I_m \sin(\omega t) \quad \Rightarrow \mathbf{I} = I_m \underline{/0^\circ}.$$

Therefore,

$$v(t) = i(t)R = RI_m \sin \omega t,$$

which gives the phasor representation of the voltage to be

$$\mathbf{V} = RI_m \underline{/0^\circ}.$$

The phase angle of both voltage and current waveforms is 0°, and they are said to be in-phase.

6.5.2 Inductor

Assume that the current through the inductor is given by

$$i(t) = I_m \sin \omega t \quad \Rightarrow \mathbf{I} = I_m \underline{/0^\circ}.$$

We have

$$\begin{aligned} v_L(t) &= L\frac{di(t)}{dt} \\ &= \omega L I_m \cos \omega t \\ &= \omega L I_m \sin(\omega t + 90^\circ). \end{aligned}$$

This gives the phasor representation of the voltage to be

$$\mathbf{V}_L = \omega L I_m \underline{/90^\circ},$$

indicating that the voltage leads the current by 90° OR that the current lags the voltage by 90°.

6.5.3 Capacitor

Assume that the voltage across the capacitor is given by

$$v_c(t) = V_m \sin \omega t \quad \Rightarrow \mathbf{V}_c = V_m \underline{/0^\circ}.$$

We have

$$\begin{aligned} i(t) &= C\frac{dv_c(t)}{dt} \\ &= \omega C V_m \cos \omega t \\ &= \omega C V_m \sin(\omega t + 90^\circ), \end{aligned}$$

which gives the phasor representation of the current to be

$$\mathbf{I} = \omega C V_m \underline{/90^\circ}.$$

Here, the current leads the voltage by 90° OR the voltage lags the current by 90°. Note the impedance of a circuit element in an AC circuit is the ratio of the voltage phasor to the current phasor.

6.6 Reactance

Reactance is the imaginary component of impedance. Since only inductors and capacitors have a complex impedance, they are the only two that have a reactance.

6.6.1 Inductor

For an inductor, the impedance is given by

$$Z_L = j\omega L.$$

This gives the reactance of the inductor to be

$$X_L = \omega L = 2\pi f L\Omega,$$

which we can use to define the impedance to be

$$Z_L = j\omega L = jX_L.$$

As frequency increases, the inductive reactance also increases. In a DC circuit, an inductor acts as a short circuit, which transitions to an open circuit at high frequencies.

6.6.2 Capacitor

The impedance of a capacitor is given by

$$Z_C = \frac{1}{j\omega C},$$

which gives the reactance

$$X_C = \frac{1}{wC} = \frac{1}{2\pi f C}\Omega.$$

In terms of reactance, the impedance of a capacitor is

$$Z_C = \frac{1}{j\omega C} = -jX_c.$$

As frequency increases, the reactance decreases. In a purely DC circuit, it acts as an open circuit, and as a short circuit at high frequencies.

6.7 Impedance

6.7.1 RC Circuit

Impedance in a RC (resistor and capacitor in series) is composed of resistance and reactance. The impedance is given by

$$\begin{aligned} Z &= R + \frac{1}{j\omega C} \\ &= R - \frac{j}{\omega C} = R - jX_C. \end{aligned}$$

We can represent this in phasor form by finding the magnitude,

$$\begin{aligned} |Z| &= \sqrt{R^2 + \frac{-1}{\omega C}^2} \\ &= \sqrt{R^2 + X_C^2}, \end{aligned}$$

and the angle,

$$\begin{aligned} \theta &= \tan^{-1}\left(\frac{-1}{\omega C R}\right) \\ &= \tan^{-1}\left(-\frac{X_C}{R}\right). \end{aligned}$$

Here, we take the current to be the reference, however for a parallel RC circuit, the voltage is taken as the reference.

In parallel, we can write the following equations

$$\begin{aligned}\mathbf{I}_T &= \mathbf{I}_R + j\mathbf{I}_C \\ |I_T| &= \sqrt{I_R^2 + I_C^2} \\ \tan\theta &= \frac{\mathbf{I}_C}{\mathbf{I}_R} \\ \mathbf{I}_T &= \sqrt{I_R^2 + I_C^2}\ \underline{/0^\circ}.\end{aligned}$$

6.7.2 RL Circuit

Impedance in an RL (resistor and inductor in series) circuit is composed of resistance and reactance. The impedance is given by

$$\begin{aligned}Z &= R + j\omega L \\ &= R + jX_L,\end{aligned}$$

which can be turned into a phasor

$$\begin{aligned}|Z| &= \sqrt{R^2 + (\omega L)^2} \\ &= \sqrt{R^2 X_L^2} \\ \theta &= \tan^{-1}\left(\frac{\omega L}{R}\right) \\ &= \tan^{-1}\left(\frac{X_L}{R}\right).\end{aligned}$$

If the RL circuit was instead in parallel, we can write the following equations

$$\begin{aligned}\mathbf{I}_T &= \mathbf{I}_R + \mathbf{I}_L \\ |I_T| &= \sqrt{I_R^2 + I_L^2} \\ \tan\theta &= \frac{\mathbf{I}_L}{\mathbf{I}_R} \\ \mathbf{I}_T &= \sqrt{I_R^2 + I_L^2}\ \underline{/0^\circ}.\end{aligned}$$

6.8 Current and Voltage Division

Current division occurs when two components are connected in parallel. Given two components with impedances Z_1 and Z_2, the current through each is given by

$$\begin{aligned}I_1 &= \frac{Z_2}{Z_1 + Z_2} I \\ I_2 &= \frac{Z_1}{Z_1 + Z_2}.\end{aligned}$$

If the components were instead in series, they would divide the voltage, giving

$$\begin{aligned}V_1 &= \frac{Z_1}{Z_1 + Z_2} V \\ V_2 &= \frac{Z_2}{Z_1 + Z_2} V.\end{aligned}$$

7 AC Circuit Theorems

7.1 Impedance, Admittance, and Susceptance

The reciprocal of resistance is known as conductance (G), and is expressed in siemens (S), or Ω^{-1} (note that this is only if the reactance is zero). It is given by

$$G = \frac{1}{R}.$$

Similarly, the reciprocal of impedance is known as admittance (Y), and is also in units of siemens or Ω^{-1}. It is given by

$$Y = \frac{1}{Z} = G + jB,$$

where the imaginary part B is called the susceptance, and can be either capacitive B_C or inductive B_I. Admittance is used in parallel circuit analysis, as it is equivalent to the same circuit in parallel in terms of impedance, i.e. Which we can see by plugging in our variables

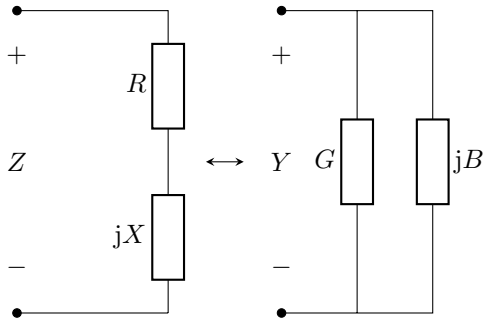


Figure 1: Transformation between impedance and admittance.

$$\begin{aligned} Y &= \frac{1}{Z} \\ &= \frac{1}{R + jX} \\ &= \frac{1}{R + jX}\frac{R - jX}{R - jX} \\ &= \frac{R}{R^2 + X^2} - j\frac{X}{R^2 + X^2} \\ &= G + jB. \end{aligned}$$

This gives the relationship between resistance, reactance, conductance, and susceptance to be

Component	Impedance	Admittance
Real	$R = \frac{G}{G^2+B^2}$	$G = \frac{R}{R^2+X^2}$
Imaginary	$X = -\frac{B}{G^2+B^2}$	$B = -\frac{X}{R^2+X^2}$

Note that $G = \frac{1}{R} \iff X = 0$. We can describe the basic elements in terms of impedance and admittance

Element	Impedance	Admittance
R	$Z_R = R$	$Y_R = \frac{1}{R}$
L	$Z_L = \mathrm{j}\omega L$	$Y_L = \frac{1}{\mathrm{j}\omega L}$
C	$Z_c = \frac{1}{\mathrm{j}\omega L}$	$Y_C = \mathrm{j}\omega L$

7.2 Delta-Wye Transformation

A Delta Δ network can be converted to a Wye Y (or Star network), and vice-versa. This is normally used in three-phase AC circuits. To transform Delta to Wye, each impedance in the Wye network is the product of the impedance in the two adjacent Delta branches, divided by the sum of the three Delta impedances. To transform back, each impedance in the Delta network is the sum of all possible products of Wye impedances taken two at a time, divided by the opposite Wye impedance.

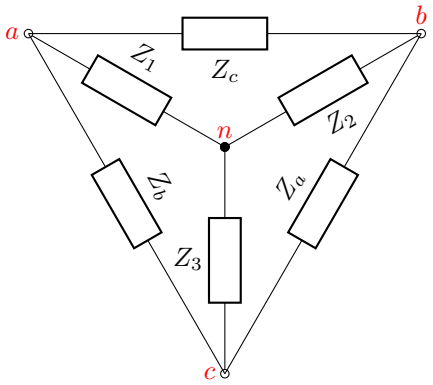


Figure 2: A Delta-Wye $(\Delta - Y)$ network.

From the figure, we can construct the relationship between Delta and Wye. Starting with transforming Delta to Wye,

$$
\begin{aligned}
Z_1 &= \frac{Z_b Z_c}{Z_a + Z_b + Z_c} \\
Z_2 &= \frac{Z_c Z_a}{Z_a + Z_b + Z_c} \\
Z_3 &= \frac{Z_a Z_b}{Z_a + Z_b + Z_c}.
\end{aligned}
$$

And to transform back,

$$
\begin{aligned}
Z_a &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1} \\
Z_b &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2} \\
Z_c &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}.
\end{aligned}
$$

A Delta or Wye circuit is balanced if it has equal impedances (load) in all three branches, i.e.

$$
\begin{aligned}
Z_Y &= Z_1 = Z_2 = Z_3 \\
&= \frac{1}{3} Z_\Delta, \\
Z_\Delta &= Z_a = Z_b = Z_c \\
&= 3 Z_Y.
\end{aligned}
$$

Since all that matters is the connection between the resistors, different arrangements of nodes and wires can still be a Delta/Wye circuit. For a Delta network, it is equivalent to a Pi network, where one resistor is at the top, and either side the other two resistors extend vertically down. Similarly, a T-network is equivalent to a Wye network, with a single resistor connect between two horizontal resistors extending vertically down. These networks are commonly used in power transmission line models.

7.3 Voltage and Current Sources

Sources can be either dependant or independent. A dependent source is controlled by another voltage or current elsewhere in the circuit, for example it might supply 0.5 times the current through a resistor.

7.3.1 Ideal Sources

An ideal voltage source has no internal resistance, and it cannot exist in nature. Practically, they always have a small amount of internal resistance, however this is usually less than 1Ω. This internal resistance appears in series with an ideal voltage source.

An ideal current source has an infinite internal resistance, and it also cannot exist in nature. Similar to a voltage source, practical sources have a very high (but not infinite) internal resistance, on the order of mega Ohms. This internal resistance is always in parallel with an ideal current source. In the real world, current sources are rarely created using a battery and resistor in series, rather they are typically constructed using transistors.

7.4 Superposition

Superposition finding the total current or voltage in a circuit by considering only one source at a time. The general process is

1. Retain one independent source at a time in the circuit and replace all other sources with their internal impedances, i.e. a voltage source should be replaced with a short circuit, a current source should be replaced with an open circuit.
2. Determine the output (current or voltage) due to the single source acting alone using any circuit analysis methods.
3. Repeat the previous two steps for each of the other independent sources.
4. Find the total contribution by adding algebraically all the contributions due to the independent sources.

7.5 Transformation Between Voltage and Current Sources

Source transformation allows us to consider a voltage source in series with an impedance the same as a current source in parallel with an impedance, and vice-versa. The impedance in both situations remains the same, however the voltage and current is given by Ohm's law

$$V_S = Z_S I_S \iff I_S = \frac{V_S}{Z_S}.$$

This can also be used for Thevenin's and Norton's equivalent circuits, which is the same as the DC case.

- Remove all the independent sources
- Assume a voltage or current source connected across the $a-b$ terminals, generally assumed to be $1\angle 0°$
- Calculate the voltage or current, and hence $Z_{Th} = \frac{v_o}{i_o}$

8 AC Circuit Analysis

8.1 Power Transfer

Applying an AC signal to a circuit containing resistive and reactive elements results in power transfer from the source to the circuit. Consider the case that the load is purely resistive. This gives our voltage and current to be

$$v(t) = V_m \cos \omega t, \quad i(t) = I_m \cos \omega t.$$

Multiplying gives the instantaneous power for the circuit as

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= V_m I_m \cos^2 \omega t, \end{aligned}$$

which is always positive, and so the energy flows continually in the direction from the source to the load. Now, consider the case where the load is purely inductive. Our voltage and current is now

$$v(t) = V_m \cos \omega t, \quad i(t) = \frac{v(t)}{Z} = I_m \sin \omega t.$$

This gives the instantaneous power to be

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= V_m I_m \cos \omega t \sin \omega t \\ &= \frac{V_m I_m}{2} \sin^2 \omega t, \end{aligned}$$

which gives the average power to be zero. Half the time the power is positive, where the inductor is storing the energy in its magnetic field, and the other half of the time it is returning that energy to the source. That means that in this case, the reactive power flows from the source to the load. Finally, considering a purely capacitive load, we have

$$v(t) = V_m \cos \omega t, \quad i(t) = \frac{v(t)}{Z} = -I_m \sin \omega t.$$

The instantaneous power is

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= -V_m I_m \cos \omega t \sin \omega t \\ &= -\frac{V_m I_m}{2} \sin^2 \omega t. \end{aligned}$$

The power of the capacitive load is the same as the inductive load, except with the opposite sign. For this, we say that the reactive power flows from the load, and that reactive power is positive for an inductance and negative for a capacitance.

8.2 AC Power

If the load contains both inductance and capacitance with reactive powers of equal magnitude, the reactive powers cancel. The power flows back and forth to inductances and capacitances is called reactive power. Even though no average power is consumed by a pure energy-storage element, reactive power is still important. It is possible to have loads composed of energy-storage elements that draw large currents that require heavy-duty wiring, even though little average power is consumed. This means that electric power delivery companies charge their (industrial) customers for reactive power (at a lower rate) as well as for the total energy delivered.

8.2.1 Power Calculation

Consider the voltage, current, and power for a general RLC load, for which the phase θ can be any value from $-90°$ to $+90°$. The voltage and current are therefore given by

$$v(t) = V_m \cos \omega t, \quad i(t) I_m \cos(\omega t - \theta).$$

The instantaneous power is then

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= V_m I_m \cos\omega t \cos(\omega t - \theta) \\ &= V_m I_m \cos\omega t (\cos\omega t \cos\theta + \sin\omega t \sin\theta) \\ &= V_m I_m \cos\theta \cos^2\omega t + V_m I_m \sin\theta \sin\omega t \cos\omega t \\ &= V_m I_m \cos\theta \left[\frac{1+\cos 2\omega t}{2}\right] + \frac{V_m I_m}{2} \sin\theta \sin 2\omega t \\ &= \frac{V_m I_m}{2} \cos\theta [1 + \cos 2\omega t] + \frac{V_m I_m}{2} \sin\theta \sin 2\omega t. \end{aligned}$$

Notice that the term involving $\cos 2\omega t$ and $\sin\omega t$ have average values of zero, thus giving the average power to be

$$P = \frac{V_m I_m}{2} \cos\theta = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos\theta = V_{rms} I_{rms} \cos\theta.$$

The term $\cos\theta$ is called the power factor (pf)

$$pf = \cos\theta,$$

and θ is sometimes called the power angle. It is common to state that the current leads (in capacitive loads, i.e. leading power factor) or lags (in inductive loads, i.e. lagging power factor) the voltage. A typical power factor would be stated to be 90% lagging, which means that $\cos\theta = 0.9$, and the current lags the voltage.

8.2.2 Power Factor

Assuming voltage $v(t)$ as the reference, look at the current $i(t)$ relative to the voltage. If the current is lagging the voltage, i.e. lagging power factor, this implies an inductive load. Inversely, if the current is leading the voltage, i.e. leading power factor, this implies a capacitive load. Note that we assume a voltage as having zero phase. In general, the phase of the voltage may have a value other than zero. Then, θ should be taken as the phase of the voltage θ_v minus the phase of the current θ_i

$$\theta = \theta_v - \theta_i.$$

The power factor is the cosine of the angle at which the current lags the voltage.

8.2.3 Reactive Power

The power flowing back and forth between the source and the energy-storage element is the last term in the power expression above

$$p(t) = \frac{V_m I_m}{2} \sin\theta \sin 2\omega t.$$

The peak value is defined to be the reactive power Q

$$Q = \frac{V_m I_m}{2} \sin\theta = V_{rms} I_{rms} \cos\theta \text{ VAR}.$$

Note that this is in units of Voltage-Amperes Reactive, which is the same as Watts except that it is specified to be reactive power. For a purely resistive load,

$$\theta = 0, \quad \therefore Q = 0.$$

Q does not represent the flow of net energy.

8.2.4 Apparent Power

The apparent power is define as the product of the effective voltage V_{rms} and effective current I_{rms}, and has units Volt-Ampere (VA). In terms of average power P and reactive power Q, it is

$$\begin{aligned} P^2 + Q^2 &= (V_{rms} I_{rms})^2 \left(\cos^2\theta + \sin^2\theta\right) \\ &= (V_{rms} I_{rms})^2 . \end{aligned}$$

If we say that we have a 5 kW load, that means that $P = 5$ kW. If we have a 5kVA load, $V_{rms} I_{rms} = 5$kVA. If we say that a load absorbs 5 kVAR, then $Q = 5$ kVAR.

8.2.5 Power Triangle

The relationship between real power (average power), reactive power, apparent power, and the power angle can be represented by the power triangle. For an inductive load, in which case $\theta > 0$, so the reactive power Q is positive. When $\theta < 0$, the reactive power Q is negative.

8.2.6 Additional Relationships

Consider a circuit given by a voltage source and a load $|Z| \ \underline{/\theta}$. We have the following relationships

$$\begin{aligned} I = \frac{V}{Z} &= \frac{V_m \underline{/0^\circ}}{|z| \, \underline{/\theta}} = I_m \underline{/-\theta} \\ \cos\theta &= \frac{R}{|z|} \\ \sin\theta &= \frac{X}{|z|} \\ I_m &= \frac{V_m}{|z|}. \end{aligned}$$

That means we can express the real power in a couple of different ways

$$P = \frac{V_m I_m}{2}\cos\theta = \frac{V_m I_m}{2}\frac{R}{|z|} = \frac{I_m^2 R}{2} = I_{rms}^2 R.$$

Similarly,

$$Q = I_{rms}^2 X.$$

For an inductive load X is positive, whereas for a capacitive load X is negative. We can also see that

$$P = \frac{V_{Rrms}^2}{R} \quad \text{and} \quad Q = \frac{V_{Xrms}^2}{X}.$$

8.2.7 Complex Power

Consider a circuit with a voltage source and some impedance in series. The complex power, denoted S, delivered to the circuit is given by

$$S = \frac{1}{2} V \bar{I} = \frac{1}{2} \left(V_m \underline{/\theta_v}\right) \left(I_m \underline{/-\theta_i}\right) = \frac{V_m I_m}{2} \underline{/\theta_v - \theta_i} = \frac{V_m I_m}{2} \underline{/\theta}.$$

We can convert this to Cartesian form to get

$$S = \frac{1}{2} V \bar{I} = \frac{V_m I_m}{2}\cos\theta + j\frac{V_m I_m}{2}\sin\theta = P + jQ.$$

This means that once we know the complex power S, we can find the average power, reactive power, and apparent power

$$P = \mathfrak{Re}(S) = \mathfrak{Re}\left(\frac{1}{2}V\bar{I}\right)$$
$$Q = \mathfrak{Im}(S) = \mathfrak{Im}\left(\frac{1}{2}V\bar{I}\right)$$
$$\text{Apparent Power} = |S| = \left|\frac{1}{2}V\bar{I}\right|.$$

8.3 Maximum Power Transfer

For a resistive circuit, the power transfer is maximised when the load is made equal to the source resistance. Consider a circuit with voltage V_{oc}, equivalent impedance $Z_{eq} = R_{eq} + jX_{eq}$, and load impedance $Z_L = R_L + jX_L$. This gives the maximum current to be

$$I_m = \frac{|V_{oc}|}{|Z_{eq} + Z_L|}.$$

To find the maximum power transfer, we need to find suitable conditions for the load parameters R_L and X_L

$$P_L = R_L I_{rms}^2, \quad P_L = R_L\left(\frac{1}{\sqrt{2}}\right)^2 \frac{\left|V_{oc}^2\right|}{\left(R_{eq}+R_L\right)^2 + \left(X_{eq}+X_L\right)^2}.$$

On average, reactances absorb zero power, and thus any non-zero values of $\left(X_{eq}+X_L\right)^2$ would only reduce P_L. We can eliminate this term by imposing $X_L = -X_{eq}$, which gives P_L to become

$$P_L = \frac{1}{2}\left|V_{oc}\right|^2 \frac{R_L}{\left(R_{eq}+R_L\right)^2}.$$

Since P_L is minimised when $\frac{dP_L}{dR_L} = 0$, then $R_L = R_{eq}$, so for maximum power transfer,

$$R_L + jX_L = R_{eq} - jX_{eq} \Rightarrow Z_L = Z_{eq}.$$

We also have the maximum average power to be

$$P_{L(max)} = \frac{1}{2}\left|V_{oc}\right|^2 \frac{R_{eq}}{\left(R_{eq}+R_{eq}\right)^2}, \quad P_{L(max)} = \frac{1}{8}\frac{\left|V_{oc}\right|^2}{R_{eq}}.$$

8.3.1 Power Factor Correction

Large currents can flow through energy storage devices (because of inductance and capacitance, as mentioned above) without average power being delivered. So, in heavy industry, many loads are partially inductive and there is a large amount of reactive power flow, which causes higher currents in the power distribution system. As a result, the lines and transformers must have a higher rating than would be necessary with a purely resistance load. This means that companies charge based on how many energy is delivered at low power factors. This means, it is better to have loads near unity power factor, which is done by placing capacitors in parallel with inductive loads to increase the power factor.

9 AC Resonance

Resonance is the tendency of a system to oscillate at maximum amplitude at a certain frequency, known as the resonant frequency. This behaviour exists for anything that is periodic, and can be electrical, mechanical, hydraulic, acoustic, etc. Electrical resonance occurs in an electric circuit at a particular frequency when the imaginary parts of impedances of circuit elements cancel each other, i.e. the total impedance is 0, and when the input voltage and current are in phase.

9.1 Series Resonance

In a series RLC circuit, the energy flows back and forth between the capacitance and inductance, and is damped by the resistor.

Consider a series RLC circuit. The impedance seen by the source is

$$\begin{aligned} Z &= R + j\omega L + \frac{1}{j\omega C} \\ &= R + j\left(\omega L - \frac{1}{\omega C}\right) \\ &= R + jX, \quad \text{where } X = \omega L - \frac{1}{\omega C}. \end{aligned}$$

The X component is the net reactance of the circuit. Depending on the applied frequency of the source ω, there are three conditions the circuit can be in

1. $\omega L < \frac{1}{\omega C}$, i.e. $X = 0$. This is capacitive behaviour
2. $\omega L > \frac{1}{\omega C}$, i.e. $X > 0$. This is inductive behaviour
3. $\omega L = \frac{1}{\omega C}$, i.e. $X = 0$. This is a purely resistive circuit with I and V in phase with each other.

This last condition is known as the unity power-factor resonance, and occurs at a specific frequency ω_0 that makes $\omega_0 L = \frac{1}{\omega_0 C}$,

$$\omega_0^2 = \frac{1}{LC}, \quad \omega = \frac{1}{\sqrt{LC}}.$$

This is the resonant frequency of the circuit, and in terms of Hertz it is

$$f_0 = \frac{1}{2\pi\sqrt{LC}} Hz.$$

Even though the individual frequencies are not zero, at resonance frequency they cancel each other out, making the total impedance 0. At resonance, we have

$$\begin{aligned} I_0 &= \frac{V_m}{R} \\ \Rightarrow V_{L0} &= (\omega_0 L)\, L_0 = \omega_0 L \frac{V_m}{R} = \frac{\omega_0 L}{R} V_m \\ V_{C0} &= \left(\frac{1}{\omega_0 C}\right) I_0 = \frac{1}{\omega_0 C}\frac{V_m}{R} = \frac{1}{\omega_0 CR} V_m. \end{aligned}$$

At series resonance, $V_{L0} = V_{C0}$, due to equal impedance, which means

$$\frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC}.$$

By definition, the quality factor of the series RLC circuit is given by

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC},$$

which we can rearrange to get

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R\sqrt{C/L}}.$$

At resonance, $\omega = \omega_0$, the voltage across the inductor is given by

$$V_{L0} = \frac{\omega_0 L}{R} V_m = QV_m.$$

Similar, at resonance the voltage across the capacitor is given by

$$V_{C0} = \frac{1}{\omega_0 RC} V_m = QV_m.$$

The peak amplitude across the reactive components exceeds the peak amplitude of the applied signal.

9.1.1 Resonance Voltage Rise

We can express the voltage and current in terms of phasors to see clearly what is happening.

$$\begin{aligned} I &= \frac{V_m}{R}\angle 0^\circ \\ V_{R0} &= V_m \angle 0^\circ \\ V_{L0} &= QV_m \angle 90^\circ \\ V_{C0} &= QV_m \angle -90^\circ. \end{aligned}$$

From this, we can see that if $Q > 1$, then the peak voltage across the components is greater than the applied signal, which is known as resonance voltage rise. The amount of magnification depends on the quality factor. It is interesting to note that the smaller the resistance R in a series RLC circuit, the higher the value of Q, and hence greater amount of voltage rise at resonance. Taking the limit of this, we see

$$\begin{aligned} \lim_{R\to 0} Q &= \lim_{R\to 0} \frac{\omega_0 L}{R} = \lim_{R\to 0} \frac{1}{\omega C R} \\ &= \infty, \end{aligned}$$

which would result in an infinite voltage rise. This quality factor is also a measure of the frequency selectivity, with a higher Q corresponding to high selectivity, or lower bandwidth.

Bandwidth is the width of the frequency response of a circuit taken when the magnitude is $-3dB$. Bandwidth is given by

$$BW = \omega_U - \omega_L,$$

i.e. the difference between the highest frequency with magnitude greater than $-3dB$ and the lowest frequency with magnitude greater than $-3dB$. We can then express the quality factor in terms of this bandwidth as

$$Q = \frac{\omega_0}{BW} = \frac{R}{L}.$$

We can also rewrite the impedance of the circuit in terms of the quality factor and the resonant frequency

$$\begin{aligned} Z &= R + j\left(\omega L - \frac{1}{\omega C}\right) \\ &= R\left[1 + j\frac{\omega L}{R} - j\frac{1}{\omega C R}\right] \\ &= R\left[1 + jQ\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right]. \end{aligned}$$

We can do the same with the magnitude of the current

$$\begin{aligned} I_m &= \frac{V_m \angle 0^\circ}{Z} \\ &= \frac{V_m}{R} \frac{1}{\sqrt{1 + Q^2\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2}}. \end{aligned}$$

The plot of I_m vs ω for a given V_m has the same profile as the band-pass function. At resonance, the average power P dissipated in the resistance is also maximised, giving

$$P_{\max} = \left(\frac{V_m}{\sqrt{2}}\right)^2 \frac{1}{R} = \frac{V_m^2}{2R}.$$

The frequencies at which P is down to half its maximum value are the half-power frequency from the bandwidth expression, ω_L and ω_U. If $Q \gg 1$, then the half-power frequencies are given by the approximate expressions

$$\begin{aligned} f_U &\approx f_0 + \frac{BW}{2} \\ f_L &\approx f_0 - \frac{BW}{2}. \end{aligned}$$

9.2 Parallel Resonance

In a parallel resonance circuit, we find everything in terms of admittance instead of impedance. We have

$$\begin{aligned} Y &= Y_R + Y_C + Y_L \\ &= G + j\omega C + \frac{1}{j\omega L} \\ &= G + j\left(\omega C - \frac{1}{\omega L}\right). \end{aligned}$$

We expect a resonance behaviour when the susceptance ωC and $-\frac{1}{\omega L}$ cancel each other out, which (as before) is when $\omega = \omega_0$. Hence, we have

$$\omega_0 C = \frac{1}{\omega_0 L},$$

giving the resonance frequency to be

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

At resonance, the output voltage has the same phase as the input voltage, and it is purely resistive. The magnitude of the current going through the capacitor and inductor going through the inductor is given by

$$\begin{aligned} |I_C| &= \frac{V_m}{(\omega_0 C)^{-1}} = V_m \omega_0 C = I_m \left(R\omega_0 C\right), \\ |I_L| &= \frac{V_m}{\omega_0 L} = \frac{I_m R}{\omega_0 L} = I_m \left(\frac{R}{\omega_0 L}\right), \end{aligned}$$

which are equal at resonance. This gives

$$|I_L| = |I_C| \Rightarrow R\omega_0 C = \frac{R}{\omega_0 L},$$

which gives the quality factor to be

$$Q = R\omega_0 C = \frac{R}{\omega_0 L}.$$

Similar to the series RLC, we can write the admittance in terms of the resonant frequency and quality factor. We can rearrange the admittance as

$$\begin{aligned} Y &= G + j\left(\omega C - \frac{1}{\omega L}\right) \\ &= G\left[1 + j\left(\omega C R - \frac{R}{\omega L}\right)\right]. \end{aligned}$$

Rewriting, we get

$$\begin{aligned} Y &= G\left[1 + jQ\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right] \\ V &= V_m \angle 0^\circ, \text{ and } I = I_m \angle 0^\circ \\ V &= ZI = \frac{1}{Y} I = \frac{I_m \angle 0^\circ}{G\left[1 + jQ\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right]} \\ &= RI_m \frac{1}{\sqrt{1 + Q^2\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2}}. \end{aligned}$$

Even though the individual admittance Y_L and Y_C are non-zero, their sum vanishes at resonance, making the parallel combination of L and C effectively behave as an open circuit. Therefore, the average power dissipated in the resistance is

$$P_{\max} = \left(\frac{I_m}{\sqrt{2}}\right)^2 = RI_{rms}^2.$$

The frequencies at which P is down to half its maximum value are the half power frequencies ω_L and ω_U, and the half-power $BW = \omega_U - \omega_L$ such that

$$BW = \frac{1}{RC}.$$

9.3 Practical Resonant Circuit

The parallel RLC configuration is known as a tuned circuit, such as in tuned oscillators and tuned amplifiers in radio communications. Of the three components making up an RLC circuit, the inductor is the least ideal because of the winding resistance R_S.

We can write a parallel non-ideal RLC circuit as

$$Y = \frac{1}{R} + j\omega C + \frac{1}{R_S + j\omega L},$$

which gives

$$Y = \frac{1}{R} + \frac{R_S}{R_S^2 + (\omega L)^2} + j\omega\left(C - \frac{L}{R_S^2 + (\omega L)^2}\right).$$

We set the imaginary part to find the frequency at which Y is purely real,

$$C - \frac{L}{R_S^2 + (\omega L)^2} = 0$$
$$\Rightarrow \omega_0 = \sqrt{\frac{1}{LC} - \frac{R_S^2}{L^2}}.$$

As the inductor approaches being ideal (i.e. the limit $R_S \to 0$), we approach the same resonant frequency from before. With the introduction of this additional resistance, we can also describe the resonant frequency in terms of the quality factor of the coil

$$Q_{coil} = \frac{\omega_0 L}{R_S}, \quad \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}\left[\sqrt{1 - \frac{1}{Q_{coil}^2}}\right].$$

A parallel circuit has high impedance at its resonant frequency, and therefore it attenuates signals at all frequencies except the resonant frequency. Varying the capacitance of the tuned circuit changes the resonant frequency, and generally inductance is kept constant.

9.4 Series to Parallel Conversion

It is possible to select a parallel resistor R_P and inductor L_P such that, at a given frequency ω, the admittance of the parallel circuit equals that of the series circuit

$$\frac{1}{R_P} + \frac{1}{j\omega L_P} = \frac{1}{R_S + j\omega L_S}.$$

Rationalising both sides and equating the real and imaginary components yields

$$\frac{1}{R_P} - j\frac{1}{\omega L_P} = \frac{R_S}{R_S^2 + \omega^2 L_S^2} - j\frac{\omega L_S}{R_S^2 + \omega^2 L_S^2}$$
$$\therefore \frac{1}{R_P} = \frac{R_S}{R_S^2 + \omega^2 L_S^2}, \quad \Rightarrow R_P = \frac{R_S^2 + \omega^2 L_S^2}{R_S}$$
$$\therefore -\frac{1}{\omega L_P} = -\frac{\omega L_S}{R_S^2 + \omega^2 L_S^2}, \quad L_P = \frac{R_S^2 + \omega^2 L_S^2}{\omega^2 L_S}$$

Note that

$$X_P = \omega L_P = \omega \left(\frac{R_S^2 + \omega^2 L_S^2}{\omega^2 L_S} \right) = \frac{R_S^2 + (\omega L_S)^2}{\omega L_S}.$$

This means we can model the internal resistance of the inductor and the resistance of actual resistor as a single resistor, i.e.

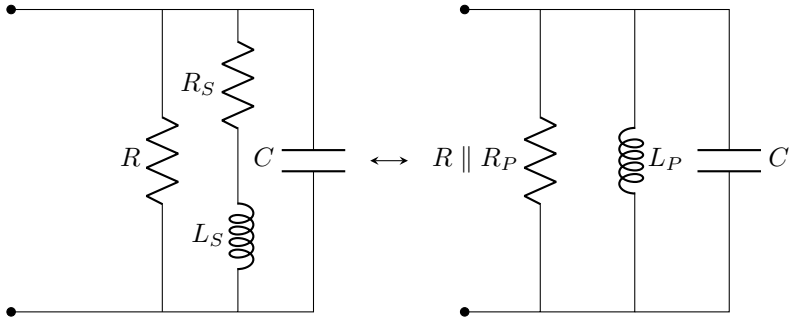


Figure 3: Transformation between non-ideal and ideal RLC circuit

In summary,

- The quality factor of the coil is given by $Q_{\text{coil}} = \frac{\omega_0 L_S}{R_S}$. Small coil resistance means large Q_{coil}
- The resonant frequency is given by $\omega_0 = \frac{1}{\sqrt{L_P C}}$
- The parallel resistance is $R_P = \frac{R_S^2 + \omega L_S^2}{R_S}$. Small R_S means large R_P
- The parallel inductance is given by $L_P = \frac{R_S^2 + \omega^2 L_S^2}{\omega^2 L_S}$. Small R_S means $L_P \approx L_S$
- The overall quality factor is $Q_{\text{circuit}} = R_P \parallel R \times \omega_0 \times C = \frac{R_P \parallel R}{\omega_0 L_P}$

10 Transient Analysis

Transient analysis looks at how a circuit or network behaves over time after a sudden change, such as when a switch opens or closes. When a circuit contains energy storage elements, it does not respond instantly to these change, and instead has a transition period before reaching a steady-state condition. There are two types of circuits

- **First-order circuits:** These contain only a single energy storage element, and can be described by a first order ODE.
- **Second-order circuits:** These contain both an inductor and a capacitor. These are described by a second order ODE.

10.1 Natural Response

10.1.1 Source-Free

A source-free first-order circuit is when the transient is the independent source being switched off. The energy stored in the elements is then released through the resistor. The natural response is the behaviour of voltages and currents after the source is switched off at $t = 0$.

For a source-free RC circuit, the natural response involves finding the voltage across the capacitor C for $t > 0$ by solving the differential equation.

For a source-free RL circuit, the natural response involves finding the current through the inductor L for $t > 0$ by solving the differential equation.

10.1.2 RC Circuit

Suppose that after a capacitor in a series RC circuit has been fully charged to V_S, the source is disconnected and the capacitor is allowed to discharge through the resistor. This then becomes a source-free circuit. This means to find the voltage across the capacitor, we have to solve

$$\frac{v}{R} + c\frac{dv}{dt} = 0 \quad \Rightarrow \frac{dv}{v} = -\frac{dt}{CR}.$$

By integrating both sides, we get

$$\begin{aligned} \ln v &= -\frac{t}{RC} + \ln A, \quad \text{where } A \text{ is some constant.} \\ \Rightarrow v(t) &= Ae^{-t/RC} \\ v(t) &= V_S e^{-t/RC}. \end{aligned}$$

The current is found as

$$i(t) = C\frac{dv}{dt} = -\frac{V_s}{R}e^{-t/RC}.$$

We define the time constant τ to be $\tau = RC$. It is the time require for the response to decay to $1/e$ (about 36.8%) of its initial value. When τ is small it decays faster, and vice-versa. The voltage starts from V_S, and decays down to 0, the current starts at $-\frac{V_S}{R}$ and decays up to 0.

10.1.3 RL Circuit

If a current source is applied to a parallel RL circuit, after a sufficient amount of time all the current will pass through the inductor, and none through the resistor. The final stored energy in the inductor is given by

$$w_L = \frac{1}{2}LI_S^2.$$

Once the load is disconnected, the inductor initially resists the change in current and induces a voltage, giving

$$i(0) = I_S \quad v(0) = -RI_S,$$

when measuring across the circuit in the same polarity as the source. We then have the response of the circuit to be

$$\begin{aligned} i(t)R + L\frac{di(t)}{dt} &= 0 \\ \Rightarrow i(t) &= I_S e^{-\frac{t}{L/R}} \\ v(t) &= L\frac{di(t)}{dt} = -RI_S e^{-\frac{t}{L/R}}. \end{aligned}$$

The current starts at the initial current I_S, and decays down to 0, and the voltage starts at $-RI_S$, and decays up to 0.

10.2 Forced Response

A forced response is the opposite of a natural response. It is where a current or voltage is suddenly applied to the circuit, and this sudden application forces the energy in the circuit to change. The most common functions for this are the unit step, unit impulse, and unit ramp. A unit impulse and unit ramp are the derivatives and integrals of the unit step, respectively.

10.2.1 RC Circuit

When a DC source is suddenly applied, it can be modelled as step function and the response of the circuit is known as the step-response. We can assume that the circuit has some initial condition $v(0) = V_0$. Applying KCL,

$$\begin{aligned} c\frac{dv(t)}{dt} + \frac{v(t) - V_S u(t)}{R} &= 0 \\ \frac{dv(t)}{dt} + \frac{v(t)}{RC} &= \frac{V_s u(t)}{CR} \\ \Rightarrow v(t) &= V_S + (V_0 - V_S)e^{-t/RC}, \quad t > 0. \end{aligned}$$

The V_S is the steady-state response, and the $(V_0 - V_S)e^{-t/RC}$ is the transient response. The complete response of a circuit is the natural response (stored energy) and force response (independent source) added together

$$v(t) = V_0 e^{-t/RC} + V_S(1 - e^{-t/RC}), \quad t > 0,$$

i.e. $v(t) = v_n(t) + v_f(t)$ (natural voltage, forced voltage).

10.2.2 RL Circuit

For an RL circuit, assume the inductor has zero initial stored energy, so that $i(0) = 0$. Connecting current source I_S causes a current, and therefore energy, buildup in the inductor. The response after this connection is given by

$$\begin{aligned} I_S &= \frac{v(t)}{R} + i(t) \\ i(t) + \frac{L}{R}\frac{di(t)}{dt} &= I_S \\ i(t) &= I_S e^{-\frac{t}{L/R}} \\ v(t) &= RI_S e^{-\frac{t}{L/R}}. \end{aligned}$$

The current starts at 0 and approaches I_S. The voltage starts at RI_S, and decays to 0.

10.3 Second-Order Circuits

A second-order circuit is one containing both inductors and capacitors. Similarly to above, the solution of the source-free RLC circuit is called the natural response of the circuit. The circuit is excited by the energy initially stored in the capacitor and inductor. The equation relating the voltage and current is given by

$$\frac{d^2 i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0, \quad Ri + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} idt = 0.$$

We can rewrite the differential form as

$$\frac{d^2 i}{dt^2} + 2\alpha\frac{di}{dt} - \omega_0^2 i = 0,$$

where $\alpha = \frac{R}{2L}$, and $\omega_0 = \sqrt{\frac{1}{LC}}$. This means that solution depend on the relative values of α and ω_0, where α is the damping factor, and ω_0 is the natural frequency of the system. We can assume the solution has the

form $i = Ae^{st}$, which we can substitute into the above to get

$$\begin{aligned} \frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} &= 0 \\ As^2e^{st} + \frac{AR}{L}se^{st} + \frac{A}{LC}e^{st} &= 0 \\ \text{or} \quad Ae^{st}\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right) &= 0 \\ s^2 + \frac{R}{L}s + \frac{1}{LC} &= 0. \end{aligned}$$

Putting back in α and ω_0, we get the characteristic equation

$$s^2 + 2\alpha s + \omega_0^2 = 0.$$

The roots of this equation are

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}, \quad s_= - \frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}},$$

or

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}.$$

These indicate two possible solutions for the current

$$i_1 = A_1e^{s_1t}, \quad i_2 = A_2e^{s_2t}.$$

A total solution is therefore a linear combination of the two solutions

$$i(t) = A_1e^{s_1t} + A_2e^{s_2t},$$

where the constants A_1 and A_2 are determined from the initial values $i(0)$ and $\frac{di(0)}{dt}$. This is natural response of a series RLC circuit. Note that we can find $\frac{di(0)}{dt}$ as

$$Ri(0) + L\frac{di(0)}{dt} + v_c(0) = 0, \quad \Rightarrow \frac{di(0)}{dt} = -\frac{1}{L}\left(v_c(0) + Ri(0)\right).$$

10.4 Initial and Final Values

The initial and final values of a second-order circuit means finding

$$\frac{dv}{dt}, \frac{di}{dt}, v(0), i(0), v(\infty), i(\infty).$$

Note that v denotes capacitor voltage, and i denotes inductor current, and that these values are always continuous, i.e.

$$v(0^-) = v(0^+), \quad i(0^-) = i(0^+).$$

Example 10.1. The switch in Fig. 4 has been closed for a long time. It is open at $t = 0$. Find the initial and final values.

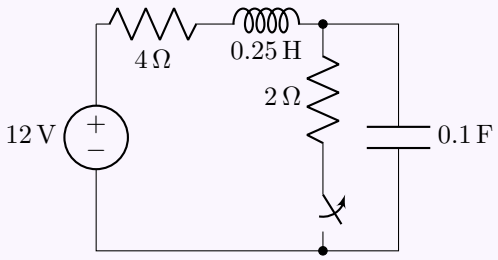


Figure 4: Series RLC circuit. The switch opens at $t = 0$

Answer:

If the switch has been closed for a long time before $t = 0$, then the inductor acts as a short circuit, and the capacitor as an open circuit. This gives

$$i(0^-) = \frac{12}{4+2} = 2\,\text{A}, \quad v(0^-) = 2i(0) = 4\,\text{V}.$$

As the inductor current and capacitor voltage cannot change abruptly,

$$i(0^+) = i(0^-) = 2\,\text{A}, \quad v(0^+) = v(0^-) = 4\,\text{V}.$$

At $t = 0^+$, the switch is open, and the equivalent circuit just becomes

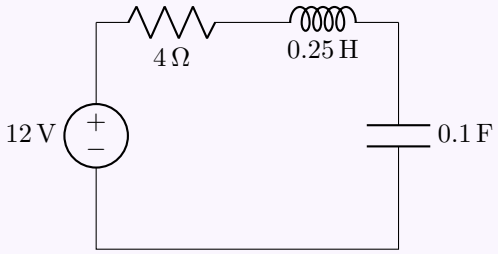


Figure 5: Series RLC circuit after the switch is closed.

The same current flows through the whole circuit, so

$$i_C(0^+) = i(0^+) = 2\,\text{A}.$$

Since $C\frac{dv}{dt} = i_C$, we have

$$\begin{aligned} \frac{dv}{dt} &= \frac{i_C}{C} \\ \frac{dv(0^+)}{dt} &= \frac{i(0^+)}{C} = \frac{2}{0.1} \\ &= 20\,\text{V}\,\text{s}^{-1}. \end{aligned}$$

Similarly, since $L\frac{di}{dt} = v_L$, we have $\frac{di}{dt} = \frac{v_L}{L}$. We can now obtain v_L by applying KVL to the loop formed by the components

$$-12 + 4i(0^+) + v_L(0^+) + v(0^+) = 0,$$

or

$$v_L(0^+) = 12 - 8 - 4 = 0.$$

Thus,

$$\frac{di(0^+)}{dt} = \frac{v_L(0^+)}{L} = \frac{0}{0.25} = 0\,\text{A}\,\text{s}^{-1}$$

For $t > 0$, the circuit undergoes transience, but as $t \to \infty$, the circuit reaches steady state again. The inductor acts like a short circuit, and the capacitor like an open circuit, and as such our circuit becomes open, and

$$i(\infty) = 0\,\text{A}, \quad v(\infty) = 12\,\text{V}.$$

Example 10.2. In the circuit in Fig. 6, calculate (a) $i_L(0^+)$, $v_C(0^+)$, $v_R(0^+)$, (b) $di_L(0^+)/dt$, $dv_C(0^+)/dt$, $dv_R(0^+)/dt$), and (c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$.

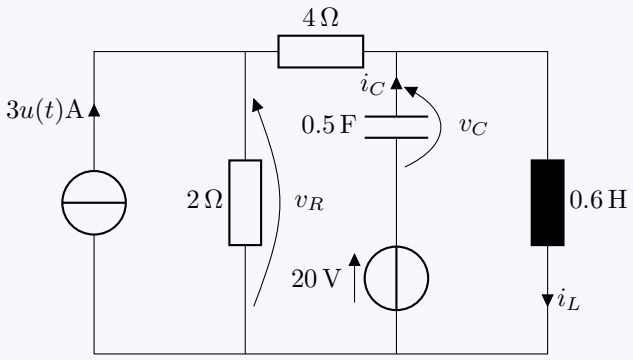


Figure 6: An RLC circuit utilising two different sources.

Answer:

(a) For $t < 0$, $3u(t) = 0$, and so this can be replaced with an open circuit. At $t = 0^-$, since the circuit has reached steady state, the inductor can be replaced by a short circuit and the capacitor with an open circuit. From this, we get

$$i_L(0^-) = 0, \quad v_R(0^-) = 0, \quad v_C(0^-) = -20\,\text{V}.$$

At $t > 0$, $3u(t) = 3$, so the circuit now is as it looks in the above figure. Since the inductor current and capacitor voltage cannot change instantaneously,

$$i_L(0^+) = i_L(0^-) = 0, \quad v_C(0^+) = v_C(0^-) = -20\,\text{V}.$$

Although we do not need to find the voltage across the 4 Ω resistor, we can use it to apply KVL and KCL. Calling it v_0, we can apply KCL on the left node to get

$$3 = \frac{v_R(0^+)}{2} + \frac{v_0(0^+)}{4},$$

and KVL to the middle mesh,

$$-v_R(0^+) + v_0(0^+) + v_C(0^+) + 20 = 0.$$

Since we know $v_C(0^+) = -20\,\text{V}$, we have

$$\begin{aligned} -v_R(0^+) + v_0(0^) - 20 + 20 = 0 \\ \Rightarrow v_R(0^+) = v_0(0^+). \end{aligned}$$

We can plug this into the KCL expression above to get

$$v_R(0^+) = v_0(0^+) = 4\,\text{V}.$$

(b) Since $L\frac{di_L}{dt} = v_L$, we get

$$\frac{di_L(0^+)}{dt} = \frac{v_L(0^+)}{L}.$$

Applying KVL now to the mesh on the right, we get

$$-20 + v_C - v_L(0^+) = 0 \quad \Rightarrow v_L(0^+) = v_C(0^+) + 20 = 0.$$

Hence,

$$\frac{di_L(0^+)}{dt} = 0.$$

Similarly, since $C\frac{dv_C}{dt} = i_C$, then $\frac{dv_C}{dt} = \frac{i_C}{C}$. Applying KCL to the node on the right of the resistor, we get

$$\frac{v_0(0^+)}{4} = i_C(0^+) + i_L(0^+).$$

Since $v_0(0^+) = 4$ and $i_L(0^+)$, $i_C(0^+) = \frac{4}{4} = 1\,\text{A}$. Then,

$$\frac{dv_C(0^+)}{dt} = \frac{i_C(0^+)}{C} = \frac{1}{0.5} = 2\,\text{V}\,\text{s}^{-1}.$$

To get $\frac{dv_R(0^+)}{dt}$, we can apply KCL to the first node again, then take the derivative of each term

$$3 = \frac{v_R}{2} + \frac{v_0}{4}$$
$$0 = 2\frac{dv_R(0^+)}{dt} + \frac{dv_0(0^+)}{dt}.$$

Doing the same by applying KVL to the middle mesh,

$$-v_R + v_C + 20 + v_0 = 0$$
$$-\frac{dv_R(0^+)}{dt} + \frac{dv_C(0^+)}{dt} + \frac{dv_0(0^+)}{dt} = 0.$$

Substituting $\frac{dv_C(0^+)}{dt} = 2$ gives

$$\frac{dv_R(0^+)}{dt} = 2 + \frac{dv_0(0^+)}{dt}.$$

Finally, using this equation and the KVL derivative, we get

$$\frac{dV_R(0^+)}{dt} = \frac{2}{3}\text{V}\,\text{s}^{-1}.$$

(c) At $t \to \infty$, the circuit reaches steady state. We this is the same as when $t < 0$, however with the addition of the current source. By current division, we get

$$i_L(\infty) = \frac{2}{2+4}3\,\text{A} = 1\,\text{A}$$
$$v_R(\infty) = \frac{4}{2+4}3\,\text{A} \times 2 = 4\,\text{V}$$
$$v_C(\infty) = -20\,\text{V}.$$

10.5 Series RLC Natural Responses

There are three types of solutions (responses) possible in a series RLC circuit

1. If $\alpha > \omega_0$, the circuit is over damped

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}, \quad \text{where } s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}.$$

Both s_1 and s_2 are real and negative. The response decays to zero.

2. If $\alpha = \omega_0$, the circuit is critically damped

$$i(t) = (A_1 + A_2 t)\, e^{\alpha t}, \quad \text{where } s_{1,2} = -\alpha.$$

The two roots are negative, real, and equal. The response reaches a maximum value at $\frac{1}{\alpha}$ before decaying to zero.

3. If $\alpha < \omega_0$, the circuit is under damped

$$i(t) = e^{-\alpha t}\left(B_1 \cos \omega_d t + B_2 \sin \omega_d t\right), \quad \text{where } \omega_d = \sqrt{\omega_0^2 - \alpha^2}.$$

Here, $B_1 = A_1 + B_2$, and $B_2 = j(A_1 - A_2)$. This response is a decaying oscillation.

10.5.1 Under-Damped Condition

This is when $\alpha < \omega_0$, and is a decaying oscillation. The roots of this solution are complex numbers, given by

$$s_1 = -\alpha + \sqrt{-\left(\omega_0^2 - \alpha^2\right)} = -\alpha + j\omega_d, \quad s_2 = -\alpha - \sqrt{-\left(\omega_0^2 - \alpha^2\right)} = -\alpha - j\omega_d,$$

where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$. The response is therefore

$$\begin{aligned} i(t) &= A_1 e^{(-\alpha - j\omega_d)t} + A_2 e^{(-\alpha + j\omega_d)t} \\ &= e^{-\alpha t}\left(A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t}\right). \end{aligned}$$

Using Euler's identities, we get

$$\begin{aligned} i(t) &= e^{-\alpha t}\left[A_1\left(\cos \omega_d t + j \sin \omega_d t\right) + A_2\left(\cos \omega_d t - j \sin \omega_d t\right)\right] \\ &= e^{-\alpha t}\left[(A_1 + A_2)\cos \omega_d t + j(A_1 - A_2)\sin \omega_d t\right] \\ &= e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t), \end{aligned}$$

where $B_1 = A_1 + A_2$ and $B_2 = j(A_1 - A_2)$.

10.6 Series RLC Forced Response

The step response is obtained by the sudden application of a DC source to an RLC circuit

$$L\frac{di}{dt} + Ri + v = V_s, \quad i = C\frac{dv}{dt}.$$

This gives the expression

$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L}\frac{dv(t)}{dt} + \frac{dv(t)}{dLC} = \frac{V_s}{LC}.$$

This is of the same form as the source-free RLC circuit equation above, with the same coefficients. The only difference is different circuit variables in the equation. The solution of the equation should have two components, the transient response v_t and the steady-state response $v_{ss}(t)$

$$v(t) = v_t(t) + v_{ss}(t).$$

The transient response (as in the first-order case) is the same as that for the source-free case (natural response). The steady-state response is the final value of $v(t)$, i.e. $v_{ss}(t) = v(\infty)$.

10.7 Parallel RLC Natural Response

Let the initial current

$$i(0) = I_0 = \frac{1}{L}\int_{-\infty}^{0} v(t)dt, \quad v(0) = V_0.$$

Applying KCL to the top node, we get

$$\frac{v}{R} + \frac{1}{L}\int_{-\infty}^{t} vdt + C\frac{dv}{dt} = 0.$$

Taking the derivative with respect to t and dividing by C,

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{RC}v = 0,$$

which is the second-order expression for a parallel RLC circuit. As before, there are three possible types of solutions for this differential equation

- If $\alpha > \omega_0$, the system is over-damped, $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$, where $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$
- If $\alpha = \omega_0$, the system is critically damped, $v(t) = (A_2 + A_1 t)\, e^{-\alpha t}$, where $s_{1,2} = -\alpha$
- If $\alpha < \omega_0$, the system is under-damped, $v(t) = e^{-\alpha t}\left(B_1 \cos\omega_d t + B_2 \sin\omega_d t\right)$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ and
$$s_{1,2} = -\alpha \pm \sqrt{-\left(\omega_0^2 - \alpha^2\right)} = -\alpha \pm j\omega_d.$$

10.8 Parallel RLC Forced Response

The step response is obtained by the sudden application of a DC source to the circuit, giving the second-order expression

$$\frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{i}{RC} = \frac{I_S}{LC}.$$

It has the same form as the equation for the source-free parallel RLC circuit, with the same coefficients. The difference is different circuit variables. The solution should have two components, the transient response $i_t(t)$, and the steady-state response $i_{ss}(t)$

$$i_(t) = \begin{cases} A_1^{s_1 t} + A_2 e^{s_2 t} & \alpha > \omega_0 \\ (A_1 + A_2 t)\, e^{-\alpha t} & \alpha = \omega_0 \\ e^{-\alpha t}\left(A_1 \cos\omega_d t + A_2 \sin\omega_d t\right) & \alpha < \omega_0 \end{cases}.$$

The steady-state response is the final value of $i(t)$

$$i_{ss}(t) = i(\infty) = I_S.$$

The values of A_1 and A_2 are obtained from the initial conditions $i(0)$ and $\frac{di(0)}{dt}$.

Part IV
Appendix

A Filters

A.1 Types of Filters

The following are idealised specifications for the most common filter functions

Ideal Low-Pass Filter

Passes low frequencies, blocks high frequencies.

$$|H(j\omega)| = \begin{cases} 1, & 0 \leq \omega \leq \omega_c \\ 0, & \omega > \omega_c \end{cases}$$

Passband: $0 \leq \omega < \omega_c$, stopband: $\omega_c < \omega \leq \infty$.

Ideal High-Pass Filter

Passes high frequencies, blocks low frequencies.

$$|H(j\omega)| = \begin{cases} 0, & 0 \leq \omega < \omega_c \\ 1, & \omega > \omega_c \end{cases}$$

Passband: $\omega_c < \omega \leq \infty$, stopband: $0 \leq \omega < \omega_c$.

Ideal Band-Pass Filter

Passes only frequencies between ω_L and ω_H.

$$|H(j\omega)| = \begin{cases} 1, & \omega_L \leq \omega \leq \omega_H \\ 0, & \text{otherwise} \end{cases}$$

Passband: $\omega_L \leq \omega \leq \omega_H$, stopbands: $\omega < \omega_L$ and $\omega > \omega_H$.

Ideal Band-Stop (Band-Reject) Filter

Rejects frequencies between ω_L and ω_H.

$$|H(j\omega)| = \begin{cases} 0, & \omega_L \leq \omega_H \\ 1, & \text{otherwise} \end{cases}$$

Passbands: $\omega < \omega_L$ and $\omega > \omega_H$, stopband: $\omega_L < \omega < \omega_H$.

B Useful Integrals

B.1 Integrals

For integer $n \neq 0$

$$\int \sin(ax)dx = -\frac{1}{a}\cos(ax)$$

$$\int \cos(ax)dx = \frac{1}{a}\sin(ax)$$

$$\int_0^T \sin(n\omega_0 t)dt = 0$$

$$\int_0^T \cos(n\omega_0 t)dt = 0$$

B.2 Integration by Parts

B.2.1 Key Formula

Key formula:

$$\int udv = uv - \int vdu.$$

$$\int xe^{ax}dx = \frac{e^{ax}}{a^2}(ax-1)$$
$$\int x^2e^{ax}dx = \frac{e^{ax}}{a^3}(a^2x^2-2ax+2)$$

B.2.2 Cosine

Key formula:

$$\int t\cos(n\omega_0 t)dt.$$

Let $u = t \Rightarrow dt$, $dv = \cos(n\omega_0 t)dt \Rightarrow v = \frac{1}{n\omega_0}\sin(n\omega_0 t)$

$$\boxed{\int t\cos(n\omega_0 t)dt = \frac{1}{n\omega_0}\sin(n\omega_0 t) - \frac{1}{(n\omega_0)^2}\cos(n\omega_0 t) + G}$$

B.2.3 Sine

Key formula:

$$\int t\sin(n\omega_0 t)dt.$$

Let $u = t \Rightarrow du = dt$, $dv = \sin(n\omega_0 t)dt \Rightarrow v = -\frac{1}{n\omega_0}\cos(n\omega_0 t)$

$$\boxed{\int t\sin(n\omega_0 t)dt = -\frac{t}{n\omega_0}\cos(n\omega_0 t) + \frac{1}{(n\omega_0)^2}\sin(n\omega_0 t) + G}$$

B.2.4 Definite Integration by Parts

For $0 \le t \le T$ and integer n

$$\int_0^T t\sin(n\omega_0 t)dt = \boxed{\frac{T}{n\omega_0}} \quad \text{and} \quad \int_0^T t\cos(n\omega_0 t)dt = \boxed{0}.$$

B.3 Definite Integrals

x^2 and trigonometric function:

$$\int_0^T x^2\cos(nx)dx = \left[\frac{x^2}{n}\sin(nx) + \frac{2x}{n^2}\cos(nx) - \frac{2}{n^3}\sin(nx)\right]_0^T$$
$$\int_0^T x^2\sin(nx)dx = \left[-\frac{x^2}{n}\cos(nx) + \frac{2x}{n^2}\sin(nx) + \frac{2}{n^3}\cos(nx)\right]_0^T.$$

x and trigonometric function:

$$\int_0^T x\cos(nx)dx = \left[\frac{x}{n}\sin(nx) + \frac{1}{n^2}\cos(nx)\right]_0^T$$
$$\int_0^T x\sin(nx)dx = \left[-\frac{x}{n}\cos(nx) + \frac{1}{n^2}\sin(nx)\right]_0^T.$$

Miscellaneous:

$$\int_0^T \cos(m\omega_0 t)\sin(n\omega_0 t)dt = 0 \quad \text{for all m and n}$$

$$\int_0^T \sin(m\omega_0 t)\sin(n\omega_0 t)dt = \begin{cases} 0 & \text{for all } m \neq n \\ \frac{T}{2} & \text{for } m = n \end{cases}$$

$$\int_0^T \cos(m\omega_0 t)\cos(n\omega_0 t)dt = \begin{cases} 0 & \text{for all } m \neq n \\ \frac{T}{2} & \text{for } m = n \end{cases}.$$

C Trigonometric Identities

$$\begin{aligned}
\sin(\alpha \pm \beta) &= \sin\alpha\cos\beta \pm \cos\alpha\sin\beta \\
\cos(\alpha \pm \beta) &= \cos\alpha\cos\beta \mp \sin\alpha\sin\beta \\
\cos(\alpha \pm 90^\circ) &= \mp\sin\alpha \\
\sin(\alpha \pm 90^\circ) &= \pm\cos\alpha \\
\cos\alpha\cos\beta &= \frac{1}{2}\cos(\alpha+\beta) + \frac{1}{2}\cos(\alpha-\beta) \\
\sin\alpha\sin\beta &= \frac{1}{2}\cos(\alpha-\beta) - \frac{1}{2}\cos(\alpha+\beta) \\
\sin\alpha\cos\beta &= \frac{1}{2}\sin(\alpha+\beta) + \frac{1}{2}\sin(\alpha-\beta) \\
\sin 2\alpha &= 2\sin\alpha\cos\alpha.
\end{aligned}$$

$$\begin{aligned}
\cos 2\alpha &= 2\cos^2\alpha - 1 \\
&= 1 - 2\sin^2\alpha \\
&= \cos^2\alpha - \sin^2\alpha \\
\sin^2\alpha &= \frac{1}{2}(1 - \cos 2\alpha) \\
\cos^2\alpha &= \frac{1}{2}(1 + \cos 2\alpha) \\
\sin\alpha &= \frac{e^{j\alpha} - e^{-j\alpha}}{2j} \\
\cos\alpha &= \frac{e^{j\alpha} + e^{-j\alpha}}{2} \\
e^{\pm j\alpha} &= \cos\alpha \pm j\sin\alpha \\
A\cos\alpha + B\sin\alpha &= \sqrt{A^2 + B^2}\cos\left(\alpha + \tan^{-1}\left(\frac{B}{A}\right)\right).
\end{aligned}$$

D Common Fourier Transform Pairs and Properties

Table 2: Common Signals and Their Fourier Transforms

Type	$f(t)$	$F(\omega)$
Impulse	$\delta(t)$	1
Constant	A	$2\pi A\delta(\omega)$
Signum	$\operatorname{sgn}(t)$	$\frac{2}{j\omega}$
Step (Heaviside)	$u(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$
Positive-time exponential	$e^{-at}u(t),\ a>0$	$\frac{1}{a+j\omega},\ a>0$
Negative-time exponential	$e^{at}u(-t),\ a>0$	$\frac{1}{a-j\omega},\ a>0$
Positive & negative exponential	e^{-a}	t
Complex exponential	$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$
Cosine	$\cos\omega_0 t$	$\pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]$
Sine	$\sin\omega_0 t$	$j\pi\left[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)\right]$

Table 3: Fourier Transform Properties

Name of Property	Function of Time	Fourier Transform
Definition	$f(t)$	$\mathcal{F}(\omega)$
Multiplication by constant	$Af(t)$	$A\mathcal{F}(\omega)$
Linearity	$af_1(t)+bf_2(t)$	$a\mathcal{F}_1(\omega)+b\mathcal{F}_2(\omega)$
Time shift	$f(t-t_0)$	$e^{-j\omega t_0}\mathcal{F}(\omega)$
Time scaling ($a>0$)	$f(at)$	$\frac{1}{a}\mathcal{F}\left(\frac{\omega}{a}\right)$
Modulation (frequency shift)	$e^{j\omega_0 t}f(t)$	$\mathcal{F}(\omega-\omega_0)$
Differentiation (time)	$\frac{d^n f(t)}{ddt^n}$	$(j\omega)^n\mathcal{F}(\omega)$
Convolution (time)	$\int_{-\infty}^{\infty} f_1(t)f_2(t-x)dx$	$\mathcal{F}_1(\omega)\mathcal{F}_2(\omega)$
Time multiplication	$t^n f(t)$	$(j)^n\frac{d^n\mathcal{F}(\omega)}{d\omega^n}$
Time reversal	$f(-t)$	$\mathcal{F}(-\omega)$
Integration	$\int_{-\infty}^{t} f(\tau)d\tau$	$\frac{\mathcal{F}(\omega)}{j\omega}+\pi\mathcal{F}(0)\delta(\omega)$

E Common Laplace Transform Pairs and Properties

Table 4: Basic Laplace Transforms

Type	Time-Domain	s-domain	Region of Convergence
Unit impulse	$\delta(t)$	1	all s
Delayed impulse	$\delta(t-\tau)$	$e^{-\tau s}$	all s
Unit step (Heaviside)	$u(t)$	$\frac{1}{s}$	$\mathfrak{Re}(s) > 0$
Delayed unit step	$u(t-\tau)$	$\frac{1}{s}e^{-\tau s}$	$\mathfrak{Re}(s) > 0$
Delayed function and step	$f(t-\tau)u(t-\tau)$	$e^{s-\tau}\mathcal{L}\{f(t)\}$	
Rectangular impulse	$u(t)-u(t-\tau)$	$\frac{1}{s}\left(1-e^{-\tau s}\right)$	$\mathfrak{Re}(s) > 0$
Ramp	$tu(t)$	$\frac{1}{s^2}$	$\mathfrak{Re}(s) > 0$
nth power ($n \in \mathbb{R}$)	$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\mathfrak{Re}(s) > 0, n > -1$
nth polynomial	$\frac{t^{n-1}}{(n-1)!}u(t)$	$\frac{1}{s^n}$	$\mathfrak{Re}(s) > 0, n > 1$
Exponential decay	$e^{-\alpha t}u(t)$	$\frac{1}{s+\alpha}$	$\mathfrak{Re}(s) > -\alpha$
nth power using frequency shift	$t^n e^{-\alpha t}$	$\frac{n!}{(s+\alpha)^{n+1}}$	
Sine	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2+\omega^2}$	$\mathfrak{Re}(s) > 0$
Cosine	$\cos(\omega t)u(t)$	$\frac{s}{s^2+\omega^2}$	$\mathfrak{Re}(s) > 0$
Exponentially decaying sine wave	$e^{-\alpha t}\sin(\omega t)u(t)$	$\frac{\omega}{(s+\alpha)^2+\omega^2}$	$\mathfrak{Re}(s) > -\alpha$
Exponentially decaying cosine wave	$e^{-\alpha t}\cos(\omega t)u(t)$	$\frac{s+\alpha}{(s+\alpha)^2+\omega^2}$	$\mathfrak{Re}(s) > -\alpha$